SPECTRAL ZETA VALUES IN THE RECURRENCE COEFFICIENTS OF RECIPROCAL SPECTRA

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ABSTRACT. For an increasing sequence $\Lambda = \{\lambda_m\}$ with $\lambda_m \sim m^p$ we study the recurrence coefficients of the measure $\sum_m \lambda_m^{-s/p}\delta_{1/\lambda_m}$. For the lattice $\lambda_m = m^p$ and every $p \neq 1$ we prove that $n^p b_n/c(p)$ has a double expansion in $1/N$ and in monomials N^{-1-jp} whose coefficients are the spectral zeta values $\zeta(-jp)$, in the natural index $N = n + \frac{\alpha}{2} - \zeta(0)$, $\alpha = (s - 1 - p)/p$. The integer layer is even in $1/N$ to all orders, the first zeta terms carry exact dressings $\varepsilon_j N^{-1-jp}(1 + g_j N^{-2})$, and the zeta layer is a Mellin–Barnes integral with kernel $\pi[\cot(\pi w p/2) - \cot(\pi w)]$ in which the zeta values act as Toda times. For Hurwitz lattices $(m + \delta)^p$ the offset shifts only the Szegő exponent of the hard edge, so N moves by δ and $\zeta(-jp)$ becomes $\zeta(-jp, 1 + \delta)$; for $(m + \delta)^p + \kappa$, $p > 1$, the pole of the spectral zeta function contributes $pR_qW(q)\pi\cot(\pi q)\, n^{-p}$. For general Weyl-type spectra we conjecture that the Jacobi coefficients are a decoupled Bessel–Szegő model flowed along the Toda hierarchy with spectral-zeta times, plus cotangent-weighted pole terms, and we test this with predictions fixed before the measures were computed.

1. INTRODUCTION

1.1. **Setting.** Let $p > 0$, $s > 1$, and let $\Lambda = \{\lambda_m\}_{m\geq 1}$ be an increasing sequence of positive numbers with $\lambda_m \sim m^p$. We study the discrete measure on the reciprocals of Λ,

$$\sigma_\Lambda := \sum_{m\geq 1} \lambda_m^{-s/p}\, \delta_{1/\lambda_m}, \tag{1.1}$$

and the coefficients of the recurrence $xP_n = P_{n+1} + a_n P_n + b_n^2 P_{n-1}$ of its monic orthogonal polynomials; a_n, b_n are the entries of the Jacobi matrix whose spectrum is $\{1/\lambda_m\} \cup \{0\}$. For the *lattice* $\lambda_m = m^p$, $\sigma_\Lambda = \sigma_{p,s} := \sum_k k^{-s}\delta_{k^{-p}}$ is the measure of the earlier papers of this series [22, 23, 24], and

$$n^p b_n \to c(p) := \frac{1}{4}\Big(\frac{\sqrt{\pi}\,\Gamma(\frac{1}{p} + 1)}{\Gamma(\frac{1}{p} + \frac{1}{2})}\Big)^p, \qquad n^p a_n \to 2c(p), \tag{1.2}$$

with rate $O(1/n)$ and an explicit $1/n$ term [23, Theorems 1.2 and 1.4]. We write $c = c(p)$ and

$$\alpha := \frac{s - 1 - p}{p} > -1, \tag{1.3}$$

the Bessel order of the hard edge at 0 in [23]. The organising object is the *node zeta function*

$$Z_\Lambda(w) := \sum_{m\geq 1} \lambda_m^{-w}, \tag{1.4}$$

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continued meromorphically from $\operatorname{Re} w > 1/p$; for the lattice $Z_\Lambda(w) = \zeta(pw)$. Two families of numbers recur:

$$W(q) := \binom{2q}{q} c^q = \frac{(4c)^q\,\Gamma(q+\frac12)}{\sqrt{\pi}\,\Gamma(q+1)}, \qquad \varepsilon_j := W(j)\,p\,\zeta(-jp) \quad (j \ge 1). \tag{1.5}$$

$W(1/p) = 1$ is the definition of c, and $\varepsilon_1 = \mathsf{k}_p := 2pc\,\zeta(-p)$ is the constant of [23, Theorem 1.4(ii)]. The question of this paper is how the expansion of $n^p b_n/c$ continues beyond the $1/n$ term, and how it depends on Λ. The answer is that its coefficients are spectral data of Λ: the values $Z_\Lambda(-j)$ and $Z_\Lambda(0)$ and the poles of Z_Λ.

1.2. **Main results.** Put $\mu_{n,j} := 2\zeta(-jp)/(jn^{jp})$ and define the *natural index*

$$N := n + \frac{\alpha}{2} - \zeta(0) = n + \frac{\alpha}{2} + \frac12. \tag{1.6}$$

Theorem A (zeta values in the Jacobi coefficients of the lattice)**.** *Let $p > 0$, $p \ne 1$, $s > 1$.*

(i) Double expansion. *For all $K, B \ge 1$,*

$$\frac{n^p b_n}{c} = \sum_{|\mathbf{r}|\le B} \mu_n^{\mathbf{r}}\,\mathfrak{D}_{\mathbf{r},K}(n) + O\big(n^{-K-1} + n^{-2-(B+1)p}\big),$$

where $\mu_n^{\mathbf{r}} = \prod_j \mu_{n,j}^{r_j}$ and the layers *$\mathfrak{D}_{\mathbf{r},K}$ are polynomials in $1/n$ of degree at most K with coefficients rational in (p,α), independent of the zeta values and computed by a finite residue recursion.*

(ii) Natural form. *In the index N, the integer layer is $(n/N)^p F_p(N)$ with $F_p(N) = 1 + \sum_{k\ge1} f_{2k}(p,\alpha^2)N^{-2k}$ even in N to all orders; the first-order zeta layers are $(n/N)^p \varepsilon_j N^{-1-jp}(1 + g_j N^{-2} + O(N^{-4}))$ for every $j \ge 1$, and the layer μ_1^2 is $(n/N)^p \varepsilon_1^2 N^{-2-2p}(\frac{3p+1}{2p} + O(N^{-2}))$.*

(iii) The coefficients.

$$\begin{aligned}\frac{n^p b_n}{c} = \Big(\frac{n}{N}\Big)^p \Big[1 + \frac{f_2}{N^2} + \frac{f_4}{N^4} + \frac{f_6}{N^6} + \frac{\varepsilon_1}{N^{1+p}}\Big(1 + \frac{g_1}{N^2} + \frac{g_{1,4}}{N^4}\Big) \\ + \sum_{j\ge2} \frac{\varepsilon_j}{N^{1+jp}}\Big(1 + \frac{g_j}{N^2}\Big) + \frac{3p+1}{2p}\,\frac{\varepsilon_1^2}{N^{2+2p}}\Big] + O\big(n^{-7} + n^{-2-3p}\big),\end{aligned} \tag{1.7}$$

the sum over the finitely many $j \ge 2$ with $1 + jp < \min(7, 2+3p)$, where

$$f_2 = \frac{p+1}{12} - \frac{(p+2)\alpha^2}{8}, \qquad g_j = A_j + B_j\alpha^2, \qquad B_j = \frac{(p+2)\big(j^2p^2 + (j+1)p + 2\big)}{8(2j-1)p},$$

$$A_j = \frac{j^2(j^2+2j-2)p^3 + 2(j+1)(2j^2-1)p^2 + (5j^2+2j-6)p + 2(j-2)}{24(2j-1)p},$$

and f_4, f_6, $g_{1,4}$ are the rational functions (4.2), (B.1), (B.2).

Parts (i) and (iii) are Theorem 4.3; the evenness in (ii) to all orders is Corollary 5.2. Not every term of (1.7) is a coefficient of b_n for every p; Table 4.1 lists which are. Three things are visible in (1.7): the shift $n \mapsto N$ absorbs the whole $1/n$ term and the $(s-1)^2$ part of the n^{-2} term; the coefficients of the zeta monomials are the values $\zeta(-jp)$, through ε_j; and those monomials carry their own even dressings. The second result explains the zeta terms: the band factor of [23] enters the first-order correction through one linear functional e_n (§3), with asymptotic series $\sum_j \varepsilon_j n^{-jp}$.

Theorem B (the zeta layer as a Mellin–Barnes integral)**.** *Let $p > 0$, $p \neq 1$, and $0 < c_0 < 1$. The band-factor functional e_n of the lattice satisfies*

$$e_n = \frac{1}{2\pi i} \int_{c_0 - i\infty}^{c_0 + i\infty} p\, W(w)\, \zeta(-wp)\, n^{-wp}\, \pi \Big[\cot \frac{\pi w p}{2} - \cot(\pi w) \Big]\, \mathrm{d}w,$$

and $e_n = \sum_{j \le J} \varepsilon_j n^{-jp} + O(n^{-(J+1)p})$ for every J. The kernel $-\pi \cot(\pi w) W(w) (4c)^{-w}$ is the Mellin transform of the truncated per-node law $(1-u)^{-1/2} \mathbf{1}_{u<1}$, whose root fixes the soft edge $4c$ of the zeros.

The third and fourth results treat the first node sets other than the lattice. In the residue recursion the exponent α plays two roles, the Bessel order ν of the hard edge and the Szegő exponent β of the weight. Separating them gives a *decoupled model* $M(\beta, \nu)$ (§5), whose integer layer is $(n/N)^p F_p(N; \nu)$ in $N = n + \frac{\beta+1}{2}$ with F_p independent of β. For Hurwitz lattices the offset acts only on β.

Theorem C (Hurwitz lattices)**.** *Let $p \neq 1$, $-1 < \delta \le 0$ ($\delta < 0$ if $p < 1$), $a = 1 + \delta$ and $1 < s < 1 + 2p$. For σ_Λ with $\lambda_k = (k+\delta)^p$, the double expansion of $n^p b_n / c$ is that of the decoupled model with $(\beta, \nu) = (\alpha + 2\delta, \alpha)$ and band-factor parameters $2\zeta(-jp, a)/(j n^{jp})$. Consequently (1.7) holds with N replaced by $N_\delta = n + \frac{\alpha}{2} - \zeta(0, a) = n + \frac{\alpha}{2} + \frac{1}{2} + \delta$ and $\zeta(-jp)$ by $\zeta(-jp, a)$, with the same f_{2k} and g_j; the integer layer and the zeta layers of weight at most two are those of the lattice in N_δ to all orders.*

For $p > 1$ the shifted lattices $\lambda_k = (k+\delta)^p + \kappa$ have a node zeta function with a pole at $q = 1 - \frac{1}{p}$ in the variable $w \mapsto Z_\Lambda(-w)$, with residue $R_q = -\kappa q / p$. It produces a term that is not a zeta value:

Theorem D (shifted Hurwitz lattices)**.** *Let $p > 1$, $-1 < \delta \le 0$, $a = 1 + \delta$, $\kappa \neq 0$ with $\kappa > -a^p$, $1 < s < 1 + 2p$, and $\lambda_k = (k+\delta)^p + \kappa$. Then*

$$\frac{n^p b_n}{c} = \sum_{0 \le k < 1+p} d_k(p, \alpha + 2\delta, \alpha)\, n^{-k} + p\, R_q\, W(q)\, \pi \cot(\pi q)\, n^{-p} + O(n^{-1-p}),$$

where $\sum_k d_k(p, \beta, \nu) n^{-k}$ is the integer layer of $M(\beta, \nu)$, and $p R_q W(q) \pi \cot(\pi q) = -2(2-p) c \kappa$.

1.3. **The spectral dictionary.** The four theorems are instances of one reading. For a node set Λ consider

$$\frac{p}{n} \cdot \frac{1}{2\pi i} \int_{\uparrow} -W(w)\, \pi \cot(\pi w)\, Z_\Lambda(-w)\, n^{-wp}\, \mathrm{d}w. \tag{1.8}$$

Its residue terms are: at the integers $w = j \ge 1$, $W(j) p Z_\Lambda(-j) n^{-1-jp}$, the zeta terms of Theorems A–C; at $w = 0$, $p Z_\Lambda(0)/n$, the $Z(0)$ part of the index shift $N = n + \frac{\alpha}{2} - Z_\Lambda(0)$; and at the poles q of $Z_\Lambda(-w)$ other than the Weyl pole $-1/p$, with residues R_q, $p R_q W(q) \pi \cot(\pi q) n^{-1-pq}$, the term of Theorem D. The cotangent is the Mellin signature of truncating the per-node law at the soft edge (Lemma 3.3, Proposition 3.4). For a general Weyl-type spectrum we conjecture (Conjecture 7.1) that $n^p b_n / c$ is the decoupled model with Szegő exponent $\alpha - 2Z_\Lambda(0) - 1$, flowed along the Toda hierarchy with times $-2Z_\Lambda(-j)/j$ corrected by the density and mass of the nodes, plus these pole terms. The conjecture is proved for the families of Theorems A–D (for Theorem D at first order in the poles), it is exact for reciprocal squared Bessel zeros (Proposition 4.4), and it was tested on nine further measures with predictions stored before the measures were computed (§7.2).

1.4. **What is new.** As far as we know (Section 8), the following are new: an asymptotic expansion of recurrence coefficients whose coefficients are spectral zeta values at non-positive arguments, with the natural index containing $-Z(0)$ and the exact dressings g_j (Theorem A); the Mellin–Barnes form of the zeta layer (Theorem B); the identification of a Hurwitz offset with a shift of the Szegő exponent at fixed Bessel order (Theorem C); the cotangent-weighted pole terms (Theorem D, Proposition 3.4); and Conjecture 7.1. The supporting tools are new settings for known mechanisms, and we say so where they are used: the constrained equilibrium problem is known for constraints of finite mass [3, 2], and we need only its infinite-mass hard-edge case (Proposition 3.1); the evenness after a half-index shift is known for varying weights [5, 18], and our reflection uses the grading argument of [18]; the Toda hierarchy for tilts is classical [27], its use to organise the coefficients of an expansion is the method of [13], and its dispersionless limit is the genus-zero mechanism of [12]. Exact special cases with explicit recurrence coefficients (Lommel and Tricomi–Carlitz polynomials) are classical; nobody we know of reads them through zeta values.

1.5. **Organisation and notation.** Section 2 recalls the frame of [23]. Section 3 contains the potential theory that fixes the band and the soft edge, and proves Theorem B. Section 4 proves Theorem A(i), (iii) by an order-K residue recursion and exact computer algebra. Section 5 introduces the decoupled model and proves the evenness, the Toda structure of the zeta layers and the index-shift law. Section 6 proves Theorems C and D (Theorems 6.1 and 6.2); Section 7 states Conjecture 7.1 with its evidence and open problems; Section 8 relates the results to the literature. Appendix A proves the equilibrium proposition and Appendix B lists closed forms and scripts. The analytic estimates behind Sections 5 and 6 are transfers of those of [23] and of Section 4; we outline them and do not reproduce them.

Throughout, $p > 0$, $c = c(p)$, and α, W, ε_j, N are as in (1.3)–(1.6); $\binom{2q}{q} = \Gamma(2q+1)/\Gamma(q+1)^2$ for complex q. Numbers obtained by fitting floating-point data are quoted with the spread of the fit over fit windows and nuisance exponents in parentheses; such a spread is not an error bound.

2. The frame of [23]

We use the notation of [23, Sections 4, 8 and 10]. After the rescaling $z = n^p x$ the polynomials oscillate on the band $[0, 4c]$, with a hard edge at 0 and a soft edge at $4c$; the weight on the band is $W_0 = \mathsf{w}_n z^\alpha (4c - z) H_n(z)$, with a constant $\mathsf{w}_n > 0$ and the *band factor* H_n, which collects the discreteness of the nodes. With $\mathsf{T} := \frac{1}{\sqrt{2}}\left(\begin{smallmatrix} 1 & 1 \\ i & -i \end{smallmatrix}\right)$ and $P_0^\infty = \mathsf{T}((z-4c)/z)^{\sigma_3/4}\mathsf{T}^{-1}$, the error matrix R of the steepest-descent analysis has jumps $I + O(1/n)$ on two small circles around 0 and $4c$, exponentially small jumps elsewhere, and

$$n^{2p} b_n^2 = (Q_1)_{12}(Q_1)_{21}, \qquad Q_1 = P_1 + \mathsf{R}_1, \tag{2.1}$$

where P_1 and R_1 are the $1/z$ coefficients at infinity of the global parametrix and of R [23, Lemmas 4.1, 4.2, Proposition 5.6]. "Mod" denotes the *model weight* $z^\alpha(4c - z)$ (H_n replaced by a constant).

The band factor is $H_n = e^{-2G}$, where G is a Cauchy integral of an explicit jump density on $(-\infty, 0)$ for $p > 1$, and on the two seams $\arg z = \pm\pi p/2$ for $0 < p < 1$ [23, Lemmas 8.1 and 10.1]. Its moments are zeta values:

Lemma 2.1 (band factor)**.** *Let $p \neq 1$ and $K \geq 1$. The k-th moment of the jump density of G is $m_k = \zeta(-kp)/(kn^{kp})$, and*

$$\log H_n(z) - \log H_n(1) = -\sum_{k \leq K} \mu_{n,k}(z^k - 1) + O(n^{-(K+1)p}), \qquad \mu_{n,k} = \frac{2\zeta(-kp)}{kn^{kp}},$$

uniformly for $|z| \leq 4c + 2$, boundary values included, with one derivative.

For $p < 1$ this is [23, Lemma 10.1(iii)–(iv)]. For $p > 1$ it is the same computation on the negative axis: expand the jump density in powers of $\exp(-a_n r^{-1/p})$, $\operatorname{Re} a_n = 2\pi n \sin(\pi/p) > 0$, integrate termwise by [10, (5.9.1)], sum by [10, (25.2.1)], and apply the reflection formula [10, (25.4.1)].

So the band factor is an exponential tilt $q_\mu(z) = \exp(-\sum_k \mu_k z^k)$ at $\mu = \mu_n$. With

$$Q_k(z) := \sum_{j=0}^{k-1} \binom{2j}{j} c^j z^{k-1-j}, \qquad M_\mu(z) := \sum_k \mu_k Q_k(z), \qquad Q_k(4c) = \tfrac{1}{2} c^{k-1} k \tbinom{2k}{k}, \tag{2.2}$$

and $\mathcal{R}(z) = (z(z-4c))^{1/2}$, tilts act on the jump data by an entire conjugation, and their first-order effect on b_n is exactly linear:

Lemma 2.2 (gauge)**.** *For the weight $z^\alpha(4c - z)q_\mu(z)$ the jump data on the two circles are $\Delta_k^\mu = G_\mu \Delta_k^{\mathrm{mod}} G_\mu^{-1}$ with $G_\mu = P_0^\infty e^{-\mathcal{R} M_\mu \sigma_3/2} (P_0^\infty)^{-1}$, which is entire in (z, μ), $\det G_\mu = 1$.*

Lemma 2.3 (linear response)**.** *For every $\mu \in \mathbb{C}^K$, the $1/n$ coefficient of $n^p b_n/c$ for the weight of Lemma 2.2 is $d_1 + cp\, M_\mu(4c)$, with $d_1 = -p(\alpha+1)/2$; it is exactly linear in μ.*

Lemma 2.2 is the formula for the Szegő function of an exponential tilt (the arcsine moments of $[0, 4c]$ are $\binom{2j}{j} c^j$). Lemma 2.3 is a direct computation: expanding G_μ at 0 and at $4c$, the residue at 0 is unchanged and the residue at $4c$ moves by a matrix quadratic in $M_\mu(4c)$, but the combination entering the $1/n$ coefficient moves by exactly $cpM_\mu(4c)$. At $\mu = \mu_n$ this gives the zeta terms $cp\,\mu_{n,j} Q_j(4c) = \varepsilon_j n^{-jp}$, and with $\mu = 0$ the $1/n$ coefficient $d_1 = -p(\alpha+1)/2$ of [23, Theorem 1.4]; since $(1 + \delta/n)^{-p} = 1 - p\delta/n + O(n^{-2})$, d_1 is the first term of $(n/N)^p$, the origin of (1.6).

3. The band, the soft edge and the kernel

At leading order the zeros of P_n, rescaled by n^p, follow the equilibrium measure of unit mass constrained by the continuum density of the nodes [21, 11, 22]. In the edge variable $y = n^p \cdot (\text{node})$ the lattice nodes have density $\sigma(y) = \frac{1}{p} y^{-1/p-1}$, of infinite mass at the hard edge 0. This section fixes the band by that problem, shows that its soft edge carries the poles of the node zeta function with the weight $\pi \cot(\pi q)$, and proves Theorem B.

3.1. The constrained equilibrium with an infinite-mass hard edge.

For constraints of finite mass with compact support and no external field, the endpoint equation and the structure of the solution below are known [3, 2] (see Section 8). We need the case of a constraint of infinite mass at a hard edge, where that theory does not apply. Let $0 < Y \leq \infty$ and let σ be a density on $(0, Y]$ with

(H1) σ positive and locally Hölder continuous on $(0, Y)$;
(H2) $y \mapsto y\sigma(y)$ strictly decreasing on $(0, Y)$;
(H3) if $Y = \infty$, $\int_1^\infty \sigma(y) \log y \, dy < \infty$.

By (H2), $\int_0 \sigma = \infty$. The *constrained equilibrium problem* is to minimise $I(\mu) = \iint \log|x-y|^{-1}\,\mathrm{d}\mu\,\mathrm{d}\mu$ over positive μ on $[0,Y]$ with $\mu([0,Y]) = 1$ and $\mu \le \sigma$; the *soft-edge condition* is

$$F(a) := \int_a^Y \sigma(y)\Big(1-\frac{a}{y}\Big)^{-1/2}\,\mathrm{d}y = 1. \tag{3.1}$$

Proposition 3.1. *Assume* (H1)–(H3). *Equation* (3.1) *has exactly one root* $a \in (0,Y)$. *With* $s(y) := \sigma(y)\sqrt{y/(y-a)}$ *and*

$$\nu(x) := \frac{1}{\pi}\sqrt{\frac{a-x}{x}}\int_a^Y \frac{s(y)}{y-x}\,\mathrm{d}y \qquad (0<x<a), \tag{3.2}$$

the measure $\mu := \nu\,\mathrm{d}x|_{[0,a]} + \sigma\,\mathrm{d}y|_{[a,Y]}$ *has unit mass,* $0 < \nu < \sigma$ *on* $(0,a)$, *its potential is constant on* $[0,a]$ *and strictly smaller on* $(a,Y]$, *and* μ *is the unique solution of the constrained equilibrium problem.*

The proof (Appendix A) is self-contained: the density is obtained from a per-point Cauchy-transform identity, and uniqueness uses only the positivity of the logarithmic energy of signed measures of mass zero, which holds without compact support [22, Lemma 3.3]. When σ is Hölder of order $> \frac12$ near a, the proof also gives the edge behaviour $\sigma - \nu = C\sqrt{a-x}\,(1+o(1))$ with an explicit $C > 0$ (Remark A.1).

Corollary 3.2 (the lattice)**.** *For* $\sigma(y) = \frac{1}{p}y^{-1/p-1}$ *on* $(0,\infty)$ *the root of* (3.1) *is* $a = 4c(p)$.

Proof. (H1)–(H3) hold, and (3.1) reads $a^{-1/p}\frac1p B(\frac1p,\frac12) = 1$, i.e. $a^{1/p} = \sqrt{\pi}\,\Gamma(\frac1p+1)/\Gamma(\frac1p+\frac12)$, which is $a = 4c(p)$ by (1.2). □

This is the lattice problem of [22, Theorem 1.3] in the edge variable; its edge $4c$ is the soft edge of the band of [23]. In node language, (3.1) says $\frac1n\sum_{u_k<1}(1-u_k)^{-1/2} \approx 1$ with $u_k = a\lambda_k/n^p$: the edge is fixed by the per-node quantity $L(u) := (1-u)^{-1/2}\mathbf{1}_{u<1}$, summed over the saturated nodes.

3.2. The soft edge carries the poles.

Lemma 3.3. *For* $\operatorname{Re} w < 0$, $\widetilde{L}(w) := \int_0^\infty u^{-w-1}L(u)\,\mathrm{d}u = B\big(-w,\frac12\big) = -\pi\cot(\pi w)\,\Gamma(w+\frac12)/(\sqrt{\pi}\,\Gamma(w+1))$. $\widetilde{L}$ *continues meromorphically to* $\mathbb{C}$, *its poles in* $\operatorname{Re} w \ge 0$ *are simple, at* $w = j \in \mathbb{N}_0$, *with* $-\operatorname{Res}_{w=j}\widetilde{L} = \binom{2j}{j}4^{-j}$, *and*

$$-(4c)^q\,\widetilde{L}(q) = W(q)\,\pi\cot(\pi q). \tag{3.3}$$

Proof. $\int_0^1 u^{-w-1}(1-u)^{-1/2}\,\mathrm{d}u = B(-w,\frac12)$ [10, (5.12.1)]; the reflection formula [10, (5.5.3)] gives $\sqrt{\pi}\Gamma(-w)/\Gamma(\frac12-w) = -\pi\cot(\pi w)\Gamma(w+\frac12)/(\sqrt{\pi}\Gamma(w+1))$, and $\Gamma(j+\frac12)/(\sqrt{\pi}j!) = \binom{2j}{j}4^{-j}$. □

The two analytic continuations $\mp i|1-u|^{-1/2}$ of $(1-u)^{-1/2}$ to $u>1$ have Mellin transforms with $-\pi e^{\pm i\pi w}/\sin(\pi w)$; their average is L, and the average of the transforms is $-\pi\cot(\pi w)$: the cotangent is the signature of truncating the per-node law at the edge. At $q \in -\frac12 - \mathbb{N}_0$ the right side of (3.3) is a limit $0\cdot\infty$, defined by the left side.

Let $\mathcal{N}$ be a C^1 counting function on $[\lambda_0,\infty)$, $\lambda_0 > 0$, $\mathcal{N}' > 0$, with $\mathcal{N}'(\lambda) = \sum_{k<K}\rho_k c_{\rho_k}\lambda^{\rho_k-1} + O(\lambda^{\rho_K-1})$ for every K, exponents $1/p = \rho_0 > \rho_1 > \cdots$, $c_{1/p} = 1$. Then $Z_{\mathcal{N}}(-w) := \int \lambda^w\,\mathrm{d}\mathcal{N}(\lambda)$ is meromorphic with poles at $q = -\rho$ and residues $R_q = -\rho c_\rho$; the Weyl pole is $q = -1/p$. Assume that no pole sits at a positive integer.

Proposition 3.4. *As $A \to 0^+$,*

$$\int_{\lambda_0}^{\infty} L(A\lambda)\,\mathrm{d}\mathcal{N}(\lambda) \sim -\sum_{q} R_q \widetilde{L}(q) A^q + \sum_{j\ge 0} \binom{2j}{j} 4^{-j} Z_{\mathcal{N}}(-j) A^j, \tag{3.4}$$

with remainder $O(A^{-\rho_K})$ after the terms with $q, j < -\rho_K$. The root of $\int L(A\lambda)\,\mathrm{d}\mathcal{N}(\lambda) = n$, written $A = 4c(1+\eta)n^{-p}$, satisfies

$$\eta = \frac{p}{n}\Big[\sum_{j\ge 0} W(j) Z_{\mathcal{N}}(-j) n^{-jp} + \sum_{q \ne -1/p} R_q W(q)\,\pi\cot(\pi q)\, n^{-pq}\Big] + O(\eta^2). \tag{3.5}$$

Proof. For one power, the substitution $u = A\lambda$ and the Taylor expansion of $(1-u)^{-1/2}$ on $[0, A\lambda_0]$ give $\int_{\lambda_0}^{1/A}(1-A\lambda)^{-1/2}\rho c_\rho \lambda^{\rho-1}\,\mathrm{d}\lambda = \rho c_\rho[A^{-\rho}B(\rho,\tfrac12) - \sum_j \binom{2j}{j}4^{-j}A^j\lambda_0^{\rho+j}/(\rho+j)]$, first for $\rho > 0$, then for $\rho \notin -\mathbb{N}$ by continuation; the first term is $-R_q\widetilde{L}(q)A^q$. The remainder of the expansion of $\mathcal{N}'$ is split at $\lambda = 1/(2A)$ and gives the integer terms up to $O(A^{-\rho_K})$, whose coefficients collect into the continued values $Z_{\mathcal{N}}(-j)$. The Weyl pole contributes $n(1+\eta)^{-1/p}$ by Corollary 3.2; by (3.3) and $\binom{2j}{j}4^{-j}(4c)^j = W(j)$ the rest at $\eta = 0$ is the bracket of (3.5), and the implicit function theorem gives (3.5). □

The values $Z_{\mathcal{N}}(-j)$ depend on the cutoff λ_0; they are not the values $Z_\Lambda(-j)$ of a node set, which enter through the band factor (Theorem 3.5). What (3.5) contributes for a node set is the pole part: the weight $W(q)\pi\cot(\pi q)$ of each non-Weyl pole. Numerically, for $\lambda_k = (k+\gamma k^{-m})^p$ at four values of (p,m) the first-order pole term of the root agrees with $pR_qW(q)\pi\cot(\pi q)$ to between 10^{-8} and 10^{-6} (script `m1_edge`).

3.3. **The band-factor functional.** For $p > 1$ the function G of the band factor has a jump $i\mathfrak{e}$ on $(-\infty, 0)$, exponentially small near 0 and tending to $\pi(1-2/p)$ at $-\infty$; for $0 < p < 1$ it jumps across the seams $\arg z = \pm\vartheta$, $\vartheta = \pi p/2$, by $\mathfrak{j}(|z|) = -\log(1 - e^{-2\pi n|z|^{-1/p}})$ [23, Lemmas 8.1, 10.1]. For f analytic near these contours put $\mathcal{M}_n[f] := \frac{1}{2\pi}\int_{-\infty}^{0}\mathfrak{e}(y)f(y)\,\mathrm{d}y$ $(p > 1)$ and $\mathcal{M}_n[f] := \frac{1}{\pi}\int_0^\infty \mathfrak{j}(r)\,\mathrm{Im}[e^{i\vartheta}f(re^{i\vartheta})]\,\mathrm{d}r$ $(p < 1)$, so that $\mathcal{M}_n[y^{-k-1}] = m_k$. Since $\sum_j Q_j(4c)y^{-j-1} = k_c(y) := y^{-2}(1-4c/y)^{-3/2}$ for $|y| > 4c$, the *band-factor functional*

$$e_n := 2cp\,\mathcal{M}_n[k_c] \tag{3.6}$$

is the untruncated version of the zeta terms $cpM_{\mu_n}(4c) = 2cp\sum_k m_kQ_k(4c)$ of Lemma 2.3.

Theorem 3.5. *Let $p > 0$, $p \ne 1$, $0 < c_0 < 1$. Then* (i) $e_n = 2cp\sum_{k\le K} m_kQ_k(4c) + O(n^{-(K+1)p})$ *for every K;* (ii) e_n *is the Mellin–Barnes integral of Theorem B;* (iii) $e_n = \sum_{j\le J}\varepsilon_j n^{-jp} + O(n^{-(J+1)p})$.

Proof. (i) Split at $|y| = 2(4c+2)$: on the far part expand k_c to K terms; the near parts are $O(e^{-n/C})$ [23, Lemmas 8.1(i), 10.1].

(ii), $p > 1$. For $\mathrm{Re}\,w > 0$ the computation of Lemma 2.1 with k replaced by w and [10, (25.4.1)] give

$$\int_0^\infty \mathfrak{e}(-r)r^{-w-1}\,\mathrm{d}r = \frac{2\pi}{w}\,\zeta(-wp)\,n^{-wp}\Big[\sin(\pi w)\cot\frac{\pi wp}{2} - \cos(\pi w)\Big].$$

Insert the Mellin–Barnes representation $(1+x)^{-3/2} = \frac{1}{2\pi i}\int_{(c_1)}\Gamma(-t)\Gamma(\frac32+t)\Gamma(\frac32)^{-1}x^t\,\mathrm{d}t$ $(x > 0$, $-\frac32 < c_1 < 0)$, the Mellin inversion of the Beta integral [10, (5.12.3)], into $r^{-1/2}(r+4c)^{-3/2} = r^{-2}(1+4c/r)^{-3/2}$, put $w = t+1$ and exchange the integrals; with $\Gamma(1-w)/w = \pi/(\sin(\pi w)\Gamma(1+w))$ the factors collect into $pW(w)\zeta(-wp)n^{-wp}\pi[\cot(\pi wp/2) - \cot(\pi w)]$. For $0 < p < 1$ the

seam density has Mellin transform $p(2\pi n)^{-wp}\Gamma(wp)\zeta(1+wp)$, the same representation applies with $\arg(4c/r)$ rotated to $\pi-\vartheta$, and the angular factor $\sin(\pi w-\frac{\pi wp}{2})/(\sin(\pi w)\sin\frac{\pi wp}{2}) = \cot\frac{\pi wp}{2}-\cot(\pi w)$ gives the same kernel.

(iii) On vertical lines the kernel decays like $e^{-\min(p,2)\pi|\operatorname{Im} w|}$ and $\zeta(-wp)W(w)$ grows polynomially, so the contour moves to the right. The poles of $-\pi\cot(\pi w)$ at $w=j$ give $\varepsilon_j n^{-jp}$; the poles of $\cot(\pi wp/2)$ at $w=2l/p$ lie on the zeros of $\zeta(-2l)$ and are removable. □

So the band factor contributes exactly the integer residues of $-\pi\cot(\pi w)$; the poles of $\zeta(-wp)$ and the point $w=0$ lie to the left of the strip (the term $p\zeta(0)$ of d_1 comes from the index normalisation of the frame, not from e_n). The residue sum is asymptotic only, since $\zeta(-jp)$ grows factorially. At $p=1$ the two cotangents coincide, which is where the constructions of [23] change. The quadrature of e_n and the integral agree to all 16 printed digits at six values of (p,n) (scripts `spec_kernel`, `m2_thmK_plt1`).

4. Proof of Theorem A: the residue recursion

We extend the analysis of [23] to every order K in $1/n$ and every order B in the band-factor parameters. Strategy: the band factor is a tilt (Lemma 2.1), tilts act on the jump data by an entire gauge (Lemma 2.2), so every coefficient of the expansion is an entire function of the tilt parameters μ with n-independent Taylor coefficients; these are computed by a finite residue recursion in exact arithmetic. The analytic part is the order-K version of estimates proved at order 1 in [23] and at order 2 in the notes behind Section 2.

4.1. **The recursion.** We use, for $p\neq 1$, the clockwise circles ∂U_0, ∂U_r around 0 and $4c$, the local maps f (hard edge) and $\tilde\varphi$ (soft edge) and the Szegő quotients ρ, ω of [23, Sections 8–10]. Let $M:=\frac{1}{\sqrt2}\left(\begin{smallmatrix}1&i\\ i&1\end{smallmatrix}\right)$, let $\circ$ be the entrywise product, let $a_k=a_k(\alpha)$, $b_k=b_k(\alpha)$ be the coefficients of the Bessel expansions [10, (10.17.1), (10.17.8)] and u_k,v_k those of the Airy expansions [10, (9.7.2)], and

$$\Xi_k := M^{-1}\Big(M\circ\begin{pmatrix}(-1)^k a_k & a_k\\ (-1)^k b_k & b_k\end{pmatrix}\Big),\qquad X_k := M^{-1}\Big(M\circ\begin{pmatrix}u_k & (-1)^k u_k\\ v_k & (-1)^k v_k\end{pmatrix}\Big).$$

Proposition 4.1 (jumps, meromorphy and small norm to every order)**.** *Let $p\neq 1$ and $K,J\geq 1$.*

(a) *On $\partial U_0\cup\partial U_r$ the jump of the error matrix is $I+\sum_{k\leq K}\Delta_k n^{-k}+O(n^{-K-1})$ uniformly, with $\Delta_k^0 = (2f^{1/2})^{-k}P_0^\infty\rho^{\sigma_3}(\sigma_3\Xi_k\sigma_3)\rho^{-\sigma_3}(P_0^\infty)^{-1}$ on ∂U_0, $\Delta_k^r = (-1/\tilde\varphi)^k P_0^\infty(-\omega)^{-\sigma_3/2}X_k(-\omega)^{\sigma_3/2}(P_0^\infty)^{-1}$ on ∂U_r, and $\Delta_k = G_{\mu_n}\Delta_k^{\mathrm{mod}}G_{\mu_n}^{-1}+O(n^{-(J+1)p})$.*

(b) *Δ_k^{mod} is meromorphic in U_0 and U_r, and $\Delta_k^\mu := G_\mu\Delta_k^{\mathrm{mod}}G_\mu^{-1}$ has Laurent coefficients that are entire in μ.*

(c) *Define near each $e\in\{0,4c\}$, $\mathsf R_{\mathrm{in}}^{(0)}:=I$ and for $m\geq 1$*

$$\mathsf F_m(\mu):=\sum_{j=1}^{m}\mathsf R_{\mathrm{in}}^{(m-j)}(\mu)\Delta_j^\mu,\qquad \mathsf R_{\mathrm{out}}^{(m)}(\mu):=\sum_e\big(\textit{principal part of }\mathsf F_m(\mu)\textit{ at }e\big),$$
$$\mathsf R_{\mathrm{in}}^{(m)}(\mu):=\mathsf R_{\mathrm{out}}^{(m)}(\mu)-\mathsf F_m(\mu),\qquad \mathsf R_1^{(m)}(\mu):=\operatorname{Res}_0\mathsf F_m(\mu)+\operatorname{Res}_{4c}\mathsf F_m(\mu).\tag{4.1}$$

Then $\mathsf R_1=\sum_{m\leq K}\mathsf R_1^{(m)}(\mu_n)n^{-m}+O(n^{-K-1}+n^{-1-(J+1)p})$.

Proof sketch. (a) For $k=1$ this is [23, Lemmas 9.1, 9.2, 10.6]; for general k one takes $K+1$ terms of the Bessel [10, (10.40.1)–(10.40.4), (10.17.5)–(10.17.12)] and Airy [10, (9.7.5)–(9.7.6)]

expansions of the local models of [17, §6], whose remainders are $O(n^{-K-1})$ uniformly since the arguments are of size n on the circles; the sign pattern $(-1)^k$ is the same in every sector because it sits on the column carrying $e^{-\xi}$. The n-dependence of the weight is Lemma 2.1 at order J, and the conjugation is Lemma 2.2. (b) For $p > 1$, Δ_k is a limit of functions analytic in the punctured disc that grow at most polynomially at the centre. For $p < 1$, where the glued hard-edge parametrix carries seam jumps, one checks that Δ_k has no jump: X_k lies in $\operatorname{span}\{I, \sigma_2\}$ for even k and in $\operatorname{span}\{\sigma_3, i\sigma_1\}$ for odd k, so conjugation by $i\sigma_2$ gives $(-1)^k$, which cancels the sign change of $\tilde{\varphi}^k$ (and likewise at the hard edge). (c) Induction on m in the small-norm equation $\mathsf{R} = I + \mathcal{C}(\mathsf{R}_-(J - I))$, with the Cauchy operator bounded uniformly in n as in [23, Propositions 9.5, 10.8]: the term of order n^{-m} is the Cauchy integral of $\mathsf{F}_m(\mu_n)$ over the clockwise circles, which by the residue theorem is $\mathsf{R}^{(m)}_{\mathrm{out}}$ outside and $\mathsf{R}^{(m)}_{\mathrm{in}}$ inside the discs. □

Define $\mathfrak{D}_K(n;\mu) := ((Q_1)_{12}(Q_1)_{21})^{1/2}/c$, expanded to order n^{-K}, with $Q_1 = P_1 + \sum_{m\le K} \mathsf{R}_1^{(m)}(\mu)n^{-m}$, and write $\mathfrak{D}_K(n;\mu) = \sum_{\mathbf{r}} \mu^{\mathbf{r}}\mathfrak{D}_{\mathbf{r},K}(n)$. A polynomial in $1/n$ is *written in natural form* when it is re-expanded in $1/N$; a layer $\mu^{\mathbf{r}}$ carries $n^{-|\mathbf{r}|_w p}$, $|\mathbf{r}|_w := \sum_j j r_j$.

Lemma 4.2 (exact algebra)**.** *The recursion* (4.1) *and the extraction* $\mathfrak{D}_K$ *give, identically in the symbols:*

(a) $\mu = 0$, $K = 6$*:* $\mathfrak{D}_{0,6}(n) = (n/N)^p[1 + f_2N^{-2} + f_4N^{-4} + f_6N^{-6} + O(N^{-7})]$*, with no* N^{-1}, N^{-3}, N^{-5} *terms;*

(b) *first order in* μ_j, $j \ge 1$ *a symbol,* $K = 4$*: the* n^{-1} *coefficient per unit* $\binom{2j}{j}\mu_j$ *is* $pj/2$ *(*$c = 1$*), and in natural form the layer is a multiple of* $N^{-1-jp}[1 + g_jN^{-2} + O(N^{-4})]$ *with no* N^{-1}*,* N^{-3} *term; the same holds at* $K = 5$ *for* $j = 1, \dots, 5$*, and for* $j = 1$ *the* N^{-4} *coefficient of the bracket is* $g_{1,4}$*;*

(c) *second order in* μ_1*,* $K = 5$*: per unit* μ_1^2 *the natural form has no* N^{-1-2p} *term, the coefficient* $p(3p+1)/2$ *at* N^{-2-2p}*, no* N^{-3-2p} *and* N^{-5-2p} *term, and* p^2h_2 *at* N^{-4-2p}*,*

with f_2, g_j *as in Theorem A,*

$$f_4 = \frac{(p+1)(6p^3+46p^2+55p+3)}{1440\,p(3p+2)} - \frac{5(p+1)(p+2)\alpha^2}{96(3p+2)} - \frac{p(p+2)^2\alpha^4}{128(3p+2)}, \tag{4.2}$$

and f_6*,* $g_{1,4}$*,* h_2 *as in* (B.1)–(B.3) *of Appendix B.*

Proof. With $c = 1$ every input is rational in (p, α): the local maps have convergent series with rational coefficients, π cancels in f, the square roots are of series with leading coefficient 1 or a perfect square, and the factors $2^{-1/2}$ occur in conjugate pairs. The recursion runs on truncated Laurent series over $\mathbb{Q}(p, \alpha)[i]$; every series carries the index of its highest exactly known coefficient and every principal part checks that it is fully known, otherwise the run stops. For (b), $\partial_{\mu_j}\Delta_k^{\mu}|_{\mu=0} = [\mathcal{G}_j, \Delta_k^{\mathrm{mod}}]$ with $\mathcal{G}_j = \mathsf{T}\left(\begin{smallmatrix} 0 & -(z-4)Q_j/2 \\ -zQ_j/2 & 0 \end{smallmatrix}\right)\mathsf{T}^{-1}$, and the Taylor data of $Q_j/\binom{2j}{j}$ at 0 and 4 are rational in j (from $\sum_j Q_j(z)t^j = t/((1-zt)\sqrt{1-4t})$), so (b) is a computation over $\mathbb{Q}(p, \alpha, j)[i]$. For (c) the gauge is taken to second order. Since μ_j enters only as $c^j\mu_j$, the ratios are independent of c. The scripts and their hashes are listed in Appendix B. □

For $j = 1$, $g_1 = (p+1)(p^2+3p-2)/(24p) + (p+2)(p^2+2p+2)\alpha^2/(8p)$.

4.2. The theorem.

Theorem 4.3. *Let* $p > 0$*,* $p \ne 1$*,* $s > 1$*, and* $K, J, B \ge 1$*.*

(i) $\dfrac{n^p b_n}{c} = \displaystyle\sum_{|\mathbf{r}|\le B} \mu_n^{\mathbf{r}} \mathfrak{D}_{\mathbf{r},K}(n) + O\big(n^{-K-1} + n^{-2-(B+1)p} + n^{-1-(J+1)p}\big)$.

(ii) $\mathfrak{D}_{0,6}$, $\mu_{n,j}\mathfrak{D}_{e_j,4}$ *and* $\mu_{n,1}^2\mathfrak{D}_{2e_1,5}$ *are, in natural form,* $(n/N)^p[1 + f_2N^{-2} + f_4N^{-4} + f_6N^{-6} + O(N^{-7})]$, $(n/N)^p\varepsilon_j N^{-1-jp}[1 + g_jN^{-2} + O(N^{-4})]$ *(for* $j = 1$*:* $1 + g_1N^{-2} + g_{1,4}N^{-4} + O(N^{-5})$*), and* $(n/N)^p\varepsilon_1^2 N^{-2-2p}[\frac{3p+1}{2p} + h_2N^{-2} + O(N^{-4})]$*, the brackets free of odd powers of* $1/N$ *to the orders shown.*

(iii) *With* $K = 6$, $B = 2$ *and* $J \ge 6/p - 1$*, expansion* (1.7) *holds.*

Proof. (i) By (2.1) and Proposition 4.1(c), each coefficient of $\mathfrak{D}_K(n;\mu)$ is a polynomial in the entries of the $\mathsf{R}_1^{(i)}(\mu)$, entire in μ and independent of n, and Taylor's theorem in μ applies coefficientwise. The n^0 coefficient is 1 for all μ and the n^{-1} coefficient is exactly linear in μ (Lemma 2.3); hence $\mathfrak{D}_{\mathbf{r},K} = O(n^{-2})$ for $|\mathbf{r}| \ge 2$, and the Taylor remainder after order B is $O(|\mu_n|^{B+1}n^{-2}) = O(n^{-2-(B+1)p})$. (ii) is Lemma 4.2 with $\mu_{n,j}\binom{2j}{j}pc^j j/2 = \varepsilon_j n^{-jp}$. (iii) Keep $\mathbf{r} = 0$ at $K = 6$, $\mathbf{r} = e_j$ for $1 + jp < \min(7, 2+3p)$ and $\mathbf{r} = 2e_1$; the other layers with $|\mathbf{r}| \ge 2$ are $O(n^{-2-3p})$, and the bracket errors of (ii) are $O(n^{-7} + n^{-2-3p})$ in each range of p. □

A term CN^{-e} of (1.7) is a coefficient of $n^p b_n/c$ exactly when $e < \min(7, 2+3p)$ (Table 4.1); the others are identities of the layers but not coefficients of b_n, because the uncomputed layers $\mu_1\mu_2$, μ_1^3 enter at n^{-2-3p}. Theorem 4.3 by itself does not prove evenness to all orders; that is Corollary 5.2. Evaluated numerically at 40 digits to order 8, the recursion also gives $f_8(3,0) = 1311857/632311515$ and $f_8(2,0) = 3899/2^{21}$ after rational recognition; the denominators $(3p+2)$, $(3p+2)^2(5p+2)$, … come from the factors $1/(\frac{1}{p} + \frac{1}{2} + k)$ of the hard-edge local map (scripts `m4_rh4`, `m4_f8_denominator`).

term	exponent e	a coefficient of $n^p b_n/c$ for
f_2N^{-2}, f_4N^{-4}, f_6N^{-6}	$2, 4, 6$	all $p \ne 1$; $p > \frac{2}{3}$; $p > \frac{4}{3}$
$\varepsilon_j N^{-1-jp}$	$1 + jp$	$(j-3)p < 1$ and $1 + jp < 7$
$\varepsilon_1 g_1 N^{-3-p}$, $\varepsilon_1 g_{1,4} N^{-5-p}$	$3 + p$, $5 + p$	$\frac{1}{2} < p < 4$; $\frac{3}{2} < p < 2$
$\varepsilon_2 g_2 N^{-3-2p}$	$3 + 2p$	$1 < p < 2$
$\frac{3p+1}{2p}\varepsilon_1^2 N^{-2-2p}$	$2 + 2p$	$p < \frac{5}{2}$

TABLE 4.1. Which terms of (1.7) are coefficients of b_n ($p \ne 1$). The terms $\varepsilon_j g_j N^{-3-jp}$ ($j \ge 3$) never are at $B = 2$. Coincident exponents merge: at $p = 1/j_0$, $\varepsilon_{j_0} N^{-2}$ merges with f_2N^{-2}, and $f_2 + \varepsilon_{j_0} = f_2 - p/12$.

4.3. **An exact anchor: Bessel zeros.** For $\mu > -1$ let $j_{\mu,k}$ be the positive zeros of J_μ and $\lambda_k := (j_{\mu,k}/\pi)^2$; for $p = s = 2$, $\mu = \frac{1}{2}$ gives the lattice k^2.

Proposition 4.4. *For* $p = s = 2$ *and* $n \ge 1$*, with* $N := n + \mu/2$*,*

$$\frac{n^2 b_n}{c(2)} = \frac{4n^2}{(2n+\mu)\sqrt{(2n+\mu)^2 - 1}} = \Big(\frac{n}{N}\Big)^2 \Big(1 - \frac{1}{4N^2}\Big)^{-1/2}.$$

Proof. The Lommel polynomials of order $\nu = \mu + 1$ satisfy $xh_n = h_{n+1} + \beta_n^2 h_{n-1}$ with $\beta_n^2 = 1/(4(n+\nu-1)(n+\nu))$, and their Jacobi operator has spectrum the closure of $\{\pm 1/j_{\nu-1,k}\}$ [26, §3]; the measure is symmetric with masses proportional to $j_{\nu-1,k}^{-2}$. The quadratic transformation of symmetric measures (as in [23, Remark 9.6]) gives $b_n^{(y)} = \beta_{2n-1}\beta_{2n}$ for the image in $y = x^2$, and $c(2) = \pi^2/16$. □

Here $\alpha = -\frac{1}{2}$, and $j_{\mu,k} = \pi(k + \frac{\mu}{2} - \frac{1}{4}) + O(1/k)$ gives $Z_\Lambda(0) = \zeta(0, \frac{\mu}{2} + \frac{3}{4}) = -\frac{\mu}{2} - \frac{1}{4}$, so $N = n + \frac{\alpha}{2} - Z_\Lambda(0)$: the natural index of the dictionary, exactly. The bare function $(1 - \frac{1}{4N^2})^{-1/2} = 1 + \frac{1}{8N^2} + \frac{3}{128N^4} + \frac{5}{1024N^6} + \cdots$ is even, and its coefficients are f_2, f_4, f_6 at $(p, \alpha) = (2, -\frac{1}{2})$. The closed form agrees with the Jacobi coefficients computed from the zeros to $1.6 \cdot 10^{-15}$ (script `spec_lommel`).

5. The natural index and the Toda structure

This section supplies three tools: the evenness of the integer layer in the natural index (which completes Theorem A(ii) and is the engine of Theorem C), the Toda structure that organises the zeta layers as flows of the integer layers, and its dispersionless limit, which gives the leading term of every zeta layer. In each case the mechanism is known in other settings and we credit it where it is used; what is new is the setting, a fixed discrete measure accumulating at a hard edge, with no large parameter other than n and with zeta values as times.

5.1. **The decoupled model.** The exponent α enters the jump data of Proposition 4.1 in two ways: as the Bessel order ν in $a_k(\nu), b_k(\nu)$, which are polynomials in ν^2, and as the Szegő exponent β in $\rho^2 = \phi^{2(\beta+1)}$ at the hard edge and in $\phi^{-2(\beta+1)}$ at the soft edge, $\phi(z) = (z^{1/2} + (z - 4c)^{1/2})/(2c^{1/2})$. Keeping them independent, the recursion and the extraction of §4.1, with the gauge of Lemma 2.2, define formal series

$$\frac{n^p\, b(n;\beta,\nu;\mu)}{c} = \sum_{\mathbf{r}} \mu^{\mathbf{r}}\, \mathfrak{D}_{\mathbf{r}}(n;\beta,\nu), \qquad B(n;\beta,\nu) := \mathfrak{D}_0(n;\beta,\nu) = \sum_{k\ge 0} d_k(p,\beta,\nu)\, n^{-k}, \tag{5.1}$$

the *decoupled model* $M(\beta,\nu)$ and its *integer layer*; for $\beta = \nu = \alpha$ they are the layers of Theorem 4.3. Each d_k is rational in (p,β,ν) and polynomial in (β,ν).

5.2. **Reflection, shift and evenness.**

Theorem 5.1. *Identically in (p,β,ν), as formal series:*

(i) (reflection) $d_k(p, -\beta - 2, \nu) = (-1)^k d_k(p,\beta,\nu)$ *for all k;*

(ii) (shift) $B(n;\beta,\nu) = (n/N)^p F_p(N;\nu)$ *with $N := n + \frac{\beta+1}{2}$ and a formal series F_p in $1/N$ that does not depend on β; equivalently, for every real γ,*

$$B(n+\gamma;\beta-2\gamma,\nu) = \Big(1 + \frac{\gamma}{n}\Big)^p B(n;\beta,\nu); \tag{5.2}$$

(iii) *for $\mathbf{r} \in \{e_1, e_2, 2e_1\}$ the series $(N/n)^{(1+|\mathbf{r}|_w)p}\mathfrak{D}_{\mathbf{r}}(n;\beta,\nu)$, re-expanded in $1/N$, does not depend on β.*

Corollary 5.2 (evenness)**.** *$F_p(N;\nu) = F_p(-N;\nu)$, and F_p depends on ν only through ν^2. For the lattice and every $p \neq 1$ the integer layer of Theorem 4.3 is, for every K, $\mathfrak{D}_{0,K}(n) = (n/N)^p[1 + \sum_{2k\le K} f_{2k}(p,\alpha^2)N^{-2k}] + O(N^{-K-1})$ with $N = n + \frac{\alpha}{2} + \frac{1}{2}$.*

Proof. By (i), $B(n;\beta,\nu) = B(-n;-\beta-2,\nu)$; the natural index of the right side is $-N$, so (ii) gives $F_p(N) = F_p(-N)$. The ν-dependence is through $a_k(\nu), b_k(\nu)$ only. □

Proof of Theorem 5.1 (sketch). (i) Since $\sigma_2\mathsf{T} = \mathsf{T}\sigma_3$, P_0^∞ and P_1 commute with σ_2. The core matrices X_k and $\sigma_3\Xi_k\sigma_3$ lie in $\mathrm{span}\{I, \sigma_2\}$ for even k and in $\mathrm{span}\{\sigma_3, i\sigma_1\}$ for odd k, so conjugation by σ_2 multiplies them by $(-1)^k$, while $\sigma_2 d^{\sigma_3}\sigma_2 = d^{-\sigma_3}$. Hence $\sigma_2\Delta_k(\beta)n^{-k}\sigma_2 = \Delta_k(-\beta-2)(-n)^{-k}$, the recursion is covariant, and $(Q_1)_{12}(Q_1)_{21}$ is invariant. This grading is the device of Kuijlaars and Tibboel [18, Lemma 4.1], who use it to remove the odd powers of $1/n$ for varying weights; here it gives a covariance under $(n,\beta) \mapsto (-n, -\beta-2)$.

(ii) No transformation of the formal recursion is known to relate $M(\beta,\nu)$ to itself (the Christoffel shift $x\sigma_{p,s}=\sigma_{p,s+p}$ moves β and ν together). Instead the degree-n Riemann–Hilbert problem Y of the lattice is steepest-descended in the frame of a non-integer index $m=n+\gamma$: put $U_m(z):=m^{pn\sigma_3}Y(zm^{-p})$ and $\tilde U:=CU_mz^{\gamma\sigma_3}C^{-1}$, $C=m^{p\gamma\sigma_3}$. The residue at the node $(m/k)^p$ acquires the factor $k^{2p\gamma}$, so the band weight has exponent $\alpha-2\gamma$; the cut of $z^{\gamma\sigma_3}$ cancels the jump $e^{-2\pi im\sigma_3}$ on $(-\infty,0)$ because the degree n is an integer; and in the hard-edge disc the lip entry is $\omega^{(\alpha-2\gamma)}e^{\mp2\pi i\gamma}e^{-4\zeta^{1/2}}$, $\zeta=m^2f$, whose band limits are $e^{\mp i\pi\alpha}$. So the m-frame problem is $M(\alpha-2\gamma,\alpha)$: Bessel order α, Szegő exponent $\alpha-2\gamma$. For $p>1$, $|\alpha|<1$ and $0\le\gamma<(1-|\alpha|)/2$ the estimates of the Hurwitz analysis of §6.2, in which the index enters only as a real scale, transfer and give $m^pb_n/c=\sum_{|\mathbf r|\le B}\mu_m^{\mathbf r}\mathfrak D_{\mathbf r,K}(m;\alpha-2\gamma,\alpha)+O(m^{-K-1}+m^{-2-(B+1)p}+m^{-1-(J+1)p})$; this transfer is written out step by step in the notes of the repository, and we do not repeat it. Now fix $p\in(1,2)\setminus\mathbb Q$, $\alpha\in(-\frac12,\frac12)$, $\gamma\in(0,\frac14)$. Both frames express n^pb_n/c as $\sum_e c_en^{-e}+o(n^{-E})$ with exponents $k+|\mathbf r|_wp$, pairwise distinct since p is irrational, so the coefficients agree; at integer exponents only $\mathbf r=0$ contributes, which is (5.2) at $\beta=\alpha$. The coefficients of the difference are rational in (p,α,γ) and vanish on an open set for infinitely many p, hence identically; $\gamma=(\nu-\beta)/2$ gives (ii), and the equivalence with (5.2) is algebra.

(iii) The same comparison at exponent $k+Jp$ gives $\sum_{|\mathbf r|_w=J}\hat\mu^{\mathbf r}[X_{\mathbf r}]_k=0$, $\hat\mu_j=2\zeta(-jp)/j$, with $X_{\mathbf r}$ the difference of the two frames. For $J=1$, $\zeta(-p)\neq0$ on $(1,2)$; for $J=2$, $\zeta(-2p)[X_{e_2}]_k+4\zeta(-p)^2[X_{2e_1}]_k=0$ with brackets rational in p, and $\zeta(-2p)/\zeta(-p)^2$ is not rational (it vanishes at the odd integers $p\ge3$). Weight ≥3 would need the linear independence of the monomials $\prod_j\zeta(-jp)^{r_j}$ over $\mathbb Q(p)$, which we do not prove. □

Exact computer algebra confirms (5.2) identically in (p,α,β,γ) to order n^{-4}, and 40-digit evaluation to order n^{-8} at 32 points including $p<1$; controls with a half shift, the wrong sign, or the Bessel order moved with β fail (scripts `m4_shift_exact`, `sigma_frame_exact`, `sigma_frame_num`). These checks confirm the algebraic consequence, not the analysis. Evenness in a half-shifted index is known for varying weights with a string equation [5, Theorem 5.2], [13, 18], and in the semiclassical Laguerre case the index and the hard-edge exponent enter through $n+\alpha/2$ and α^2 [14, Theorem 2.1]. What is new here, as far as we know, is the decoupling of the Szegő exponent from the Bessel order, the identity (5.2) for a fixed discrete measure, and its proof from a second frame at a non-integer index.

5.3. **The zeta layer as a Toda flow.** In the unscaled variable the band factor $\exp(-\sum_k\mu_{n,k}z^k)$, $z=n^px$, is $\exp(\sum_kt_kx^k)$ with n-independent

$$t_k:=-\frac{2\zeta(-kp)}{k},\tag{5.3}$$

the times of the Toda hierarchy. For a tilt $e^{\sum_j t_jx^j}\mu$ of a compactly supported measure, with Jacobi matrix $\mathcal J$ and $h_n=\int\pi_n^2$, one has $\partial_{t_j}\log h_n=(\mathcal J^j)_{nn}$, $\partial_{t_j}b_n=\frac12b_n[(\mathcal J^j)_{nn}-(\mathcal J^j)_{n-1,n-1}]$ and $\partial_{t_j}a_n=b_{n+1}(\mathcal J^j)_{n+1,n}-b_n(\mathcal J^j)_{n,n-1}$; this is classical (Moser, Flaschka; see [27, Lemma 2.1, Theorem 2.4]). Let $\mathfrak A_{\mathbf r}$ be the analogues of $\mathfrak D_{\mathbf r}$ for $n^pa_n/(2c)$, from the second moment of the same recursion and the extraction of [23], and $A:=\mathfrak A_0$. Substitute $\mu_j=-u_jn^{-jp}$ with n-independent u_j, put $\hat b:=\sum_{\mathbf r}(-u)^{\mathbf r}n^{-|\mathbf r|_wp}\mathfrak D_{\mathbf r}(n)$, $\hat a$ likewise, and let $\hat{\mathcal J}$ be the tridiagonal matrix on $\mathbb Z$ with $\hat{\mathcal J}_{ii}=2(1+\frac in)^{-p}\hat a(n+i)$, $\hat{\mathcal J}_{i,i-1}=\hat{\mathcal J}_{i-1,i}=\hat b_i:=(1+\frac in)^{-p}\hat b(n+i)$, the shift $n\mapsto n+i$ acting at fixed u and re-expanded in $1/n$.

Theorem 5.3 (Toda structure)**.** *Let $p\neq1$, $1<s<1+2p$.*

(i) *For every J and $\tau \in \mathbb{R}^J$, the tilted lattice $e^{\sum_j \tau_j x^j}\sigma_{p,s}$ satisfies Theorem 4.3(i) with the same layers and $\mu_{n,j}$ replaced by $\mu_{n,j} - \tau_j n^{-jp}$, and likewise for $n^p a_n/(2c)$ with the layers $\mathfrak{A}_{\mathbf{r}}$.*

(ii) *For every j, as identities of formal series, $\partial_{u_j}\hat{b} = \frac{1}{2}c^j n^{-jp}\hat{b}[(\hat{\mathcal{J}}^j)_{00} - (\hat{\mathcal{J}}^j)_{-1,-1}]$ and $\partial_{u_j}(2\hat{a}) = c^j n^{-jp}[\hat{b}_1(\hat{\mathcal{J}}^j)_{10} - \hat{b}_0(\hat{\mathcal{J}}^j)_{0,-1}]$.*

(iii) *The identities of (ii) hold for $M(\beta,\nu)$ identically in (p,β,ν), and they determine every $\mathfrak{D}_{\mathbf{r}}$ and $\mathfrak{A}_{\mathbf{r}}$ from the integer layers B and A.*

Outline. (i) The tilted lattice is $\sigma_{p,s,\ell}$ of [23] with $\ell(k) = \exp\sum_j \tau_j k^{-jp}$, analytic of every angle [23, Definition 1.1]; its band weight is that of the lattice times an entire polynomial tilt, which by Lemma 2.2 changes the band-factor parameters to $\mu_n - \tau n^{-p\cdot}$ and nothing else (constants depend on τ; no uniformity is needed). (ii) Write $u = \tau + t$ with t as in (5.3). By (i) the tilted b_m, a_m are polynomials in u up to $O_u(m^{-p-L})$; by the Baire category theorem these errors are uniformly bounded for large m on some open ball, on which the Toda identities, integrated along u_j, give an identity of polynomials in (u,h) up to $O(n^{-p-L})$. Since L is arbitrary, the coefficient of h is (ii); no expansion is differentiated term by term. (iii) The same argument for the Hurwitz lattices of Section 6 gives (ii) on the open set $\{(\alpha+2\delta,\alpha)\}$, and the coefficients are rational. The left side of (ii) at $u^{\gamma+e_j}$ is (γ_j+1) times a new layer, the right side involves only lower layers. The transfer of the estimates in (i) is written out in the notes of the repository. □

Exact computer algebra confirms (ii) on the layers of the recursion for $j \le 3$ to second order in the times (scripts `toda_layers`, `toda_measures`); the variant with the shift at fixed scaled time μ fails at second order. Using the Toda equations with shifted indices re-expanded in $1/n$ to generate the coefficients of an expansion is the method of [13, Section 4] for varying weights $e^{-N(x^2/2+tx^{2\nu})}$; here the measure is fixed, the times enter as $u_j n^{-jp}$ with zeta base values, and the identities come from the exact flows by a category argument rather than from a string equation. In this reading the zeta layer is not new data: it is a fixed, n-independent Toda flow of the two integer layers, with times given by the zeta values.

5.4. **The index-shift law.** Let $M(\beta,\nu)$ carry parameters $\mu_a = \hat{\mu}_a n^{-pj_a}$ with exponents $j_1,\dots,j_r$, put $\hat{\varepsilon}_a := \frac{pj_a}{2}\binom{2j_a}{j_a}c^{j_a}\hat{\mu}_a$ ($\hat{\varepsilon}_j = \varepsilon_j$ for the lattice), and expand $n^p b(n;\beta,\nu;\mu)/c = \sum_k n^{-k} b_k(\mu)$.

Theorem 5.4 (index-shift law)**.** *Let $p \ne 1$. Identically in (p,β,ν), b_k is a polynomial of degree at most k in μ, and its part of degree exactly k is the n^{-k} coefficient of $(n/x)^p$, where x is the formal solution of $x = n - \frac{1}{p}\sum_a \hat{\varepsilon}_a x^{-pj_a}$. Equivalently, in natural form the layer $\mu^{\mathbf{r}}$ is $(n/N)^p \hat{\varepsilon}^{\mathbf{r}} N^{-|\mathbf{r}|-S}[(p+1+S)_{|\mathbf{r}|-1}/(\mathbf{r}!\,p^{|\mathbf{r}|-1}) + O(N^{-1})]$, $S := p\sum_a r_a j_a$.*

For one time this gives $\varepsilon_j N^{-1-jp}$, and $\frac{3p+1}{2p}$ for μ_1^2 (Theorem 4.3(ii)): at leading order in $1/n$ at each order in the times, *the zeta times act as an implicit shift of the index. Outline.* Grade the term $\hat{\mu}^{\mathbf{r}} n^{-k-S}$ by its defect $k-|\mathbf{r}|$; the equations of Theorem 5.3(ii) are in conservation form, so every term has defect ≥ 0, and the defect-zero parts solve the dispersionless Toda hierarchy with initial data $a^0 = 2cn^{-p}$, $b^0 = cn^{-p}$. These lie on the line $a = 2b$ of a hard edge at 0, which is invariant, and on it every flow is a Hopf equation $\partial_{t_j} b = V_j(b)\partial_n b$, $V_j(b) = \frac{j}{2}\binom{2j}{j}b^j$, solved by characteristics, $b = cx^{-p}$, $x = n + \sum_j t_j V_j(cx^{-p})$; Lagrange inversion gives the coefficients. This is the genus-zero mechanism of random-matrix map enumeration ([12, Section 2.3], [13]; Lagrange inversion as in [6, Section 26.4]), and the endpoints are Riemann invariants of the dispersionless Toda hierarchy (cf. [16]); the hard edge fixes one of them at 0. What differs

here is the initial profile $b = cn^{-p}$ of a fixed measure, the reduction to the line $a = 2b$, real and pole exponents, and zeta-valued times; the mechanism itself is not new. The closed form agrees with every exact layer of the recursion to fourth order for symbolic exponents and sixth order at rational parameters, and the Toda flow of the integer layers reproduces the dressings g_j and h_2; a "dressed" index shift does not reproduce h_2, so the corrections to the leading term are dispersive (scripts `shift_toda`, `shift_exact`, `shift_dressing`). The layer coefficients are rational in the exponents, so the identities continue to non-integer exponents, in particular to the pole exponents of Theorem 6.2.

Remark 5.5 (the diagonal layer and the parity of the zeta layers). Exact computer algebra on the second moment of the recursion gives, identically in (p, β, ν), $(M/n)^p A(n; \beta, \nu) = 1 + a_2 M^{-2} + a_4 M^{-4} + O(M^{-5})$ with $M = n + \frac{\beta}{2} + 1 = N + \frac{1}{2}$, $a_2 = -\frac{p+1}{24} + \frac{(p+1)(p+2)\nu^2}{8}$, no odd terms and no β to this order (script `toda_layers`); evenness and β-independence beyond order four are supported only numerically (scripts `m4_toda`, `m4_diag`). So b_n is centred at N and a_n at $N + \frac{1}{2}$, as in [5, Theorem 5.2] for varying weights. If A is even in M, then $a_n - a_{n-1}$ is odd in N and the second differences of b_n^2 are even, so by Theorem 5.3 the first-order zeta layers are $\varepsilon_j N^{-1-jp}$ times even series and the ε_1^2 layer is N^{-2-2p} times an even series, as found in Theorem 4.3(ii); the general monomial structure is open (Problem 7.4(b)). The factor $1/(2j-1)$ in g_j matches the first variation $\binom{2j-2}{j-1}/\binom{2j}{j} = j/(2(2j-1))$ of the dispersionless symbol $\frac{1}{2\pi}\int (a + 2b\cos\theta)^j \, \mathrm{d}\theta$ off the line $a = 2b$ (heuristic).

6. Hurwitz lattices and their shifts

For $\lambda_k = (k + \delta)^p$ the node zeta function is $Z_\Lambda(w) = \zeta(pw, a)$, $a := 1 + \delta$, so the dictionary predicts the index $n + \frac{\alpha}{2} - \zeta(0, a) = n + \frac{\alpha}{2} + \frac{1}{2} + \delta$ and the values $\zeta(-jp, a)$. The mechanism is an *index transfer*: the offset does not change the Bessel order of the hard edge, only the Szegő exponent, $\alpha \mapsto \beta' = \alpha + 2\delta$, and Theorem 5.1 converts the shifted exponent into the shifted natural index.

Theorem 6.1 (Hurwitz lattices). *Let $p \neq 1$, $-1 < \delta \leq 0$ ($\delta < 0$ if $p < 1$), $a = 1 + \delta$, $1 < s < 1 + 2p$, and $\sigma^\delta := \sum_k (k + \delta)^{-s} \delta_{(k+\delta)^{-p}}$. Put $\beta' := \alpha + 2\delta$, $\mu^\delta_{n,j} := 2\zeta(-jp, a)/(jn^{jp})$, $N_\delta := n + \frac{\alpha}{2} + \frac{1}{2} + \delta$ and $\varepsilon_j(\delta) := \binom{2j}{j} p\, c^j \zeta(-jp, a)$.*

(i) *For all $K, J, B \geq 1$, $\frac{n^p b_n}{c} = \sum_{|\mathbf{r}| \leq B} (\mu^\delta_n)^{\mathbf{r}} \mathfrak{D}_{\mathbf{r},K}(n; \beta', \alpha) + O(n^{-K-1} + n^{-2-(B+1)p} + n^{-1-(J+1)p})$.*

(ii) *The integer layer is $(n/N_\delta)^p F_p(N_\delta; \alpha)$, and the layers $e_1, e_2, 2e_1$ are those of the lattice in N_δ to all orders, with $\zeta(-jp)$ replaced by $\zeta(-jp, a)$; every layer e_j is $(n/N_\delta)^p \varepsilon_j(\delta) N_\delta^{-1-jp}(1 + g_j N_\delta^{-2} + O(N_\delta^{-4}))$. In particular (1.7) holds for σ^δ with N_δ, $\varepsilon_j(\delta)$, the same $f_2, f_4, f_6, g_j, g_{1,4}$ and the same remainder.*

At $\delta = 0$ this is Theorem 4.3. Nothing is claimed for $\delta > 0$, where $t = \delta$ in the index variable $t = nz^{-1/p}$ of [23] is a zero of the interpolating function that carries no mass and the pole removal of [23, Section 5] fails, nor at $p = 1$ with $\delta \neq 0$. The natural form of the layers of weight ≥ 3 is open, as for the lattice.

The constants. The analysis produces the band-factor moments as Fourier sums,

$$m_k(\delta) = -\frac{p}{\pi}\Gamma(kp)(2\pi n)^{-kp} S_k(\delta), \qquad S_k(\delta) := \sum_{m \geq 1} \frac{\sin(\pi kp/2 - 2\pi m\delta)}{m^{1+kp}}, \tag{6.1}$$

with $\mu^{\delta}_{n,k} = 2m_k(\delta)$. We identify them by Hurwitz's formula in the form stated in [9, Theorem 1.1] (quoted there from Titchmarsh): *for* $0 < a \le 1$ *and* $\operatorname{Re} s < 0$,

$$\zeta(s,a) = \frac{2\Gamma(1-s)}{(2\pi)^{1-s}}\Big\{\sin\frac{\pi s}{2}\sum_{m\ge1}\frac{\cos(2\pi ma)}{m^{1-s}} + \cos\frac{\pi s}{2}\sum_{m\ge1}\frac{\sin(2\pi ma)}{m^{1-s}}\Big\}.$$

At $s=-kp$ ($\operatorname{Re} s<0$) and $a=1+\delta\in(0,1]$ the bracket is $-S_k(\delta)$, and $\zeta(-kp,a) = kn^{kp}m_k(\delta)$.

6.1. **The index transfer.** In $t = nz^{-1/p}$ the Hurwitz nodes sit at $t\in a+\mathbb{N}_0$ with the masses of the lattice, so the weight on the band, its power z^{α} and the Bessel order are unchanged; only the phase changes, $\pi t \mapsto \pi(t-\delta)$. Interpolating with $\sin\pi(t-\delta)$ and $1/\Gamma(t-\delta)$, the outer scalar of [23] is built from $j_\delta(t) = j(t-\delta) + (t-\delta-\frac12)\operatorname{Log}(1-\delta/t) + \delta$, j the Binet function, and behaves like $C_\delta(-z)^{\delta}$ at $z=\infty$. The gauge $F^{\natural} := F_\delta(-z)^{-\delta}$ restores the normalisation at infinity and moves $(-z)^{-2\delta}$ into the band weight: the Szegő exponent becomes $\beta' = \alpha+2\delta = \alpha - 2Z_\Lambda(0) - 1$. On the lips in the hard-edge disc the gauge leaves the phase $e^{\pm2\pi i\delta}$, so the lip entry is $\omega^{(\beta')}e^{\pm2\pi i\delta}e^{-4\zeta^{1/2}}$, with band limits $e^{\mp i\pi\alpha}$: a Bessel model of order α glued to the Szegő function of $x^{\beta'}$, the decoupled model $M(\beta',\alpha)$. The hard-edge comparison is made in this α-*frame*, with β' entering only through a quotient $(-z)^{-\delta}D_{x^{2\delta}}$ that is bounded above and below near 0; a frame with Bessel order β' would fail once $|\beta'|\ge1$.

6.2. **Outline of the proof.** The proof follows [23, Sections 5–10] and Section 4: (H-a) interpolation with $t\mapsto t-\delta$; (H-b) bounds on the outer scalar, reduced by the shift identity for j_δ to the Binet bounds of [23], and the a-priori bound near 0 with the lattice exponent $|\alpha|/2$; (H-c) the moments (6.1), by [10, (5.9.1), (5.2.1), (25.2.1)], uniformly in δ, so that Lemma 2.2 holds with μ^{δ}_n; (H-d) the index transfer above, exact algebra on the jumps; (H-e) the hard-edge comparison in the α-frame and Proposition 4.1 for $M(\beta',\alpha)$ (for $p<1$ the glued hard-edge parametrix of [23, Section 10] is rebuilt with the gauged weight), with Lemma 2.3 for $M(\beta,\nu)$, $d_1 = -p(\beta+1)/2$ (exact computer algebra, script `m3a_e37_decoupled`); (H-f) Theorem 5.1 at $(\beta,\nu)=(\alpha+2\delta,\alpha)$, and the exact β-free dressings g_j of the decoupled model (scripts `m4_shift_exact`, `m3a_g5_exact`). Part (i) is (H-a)–(H-e) followed by the proof of Theorem 4.3(i); part (ii) is (i) with (H-f). The estimates of (H-a)–(H-e) are written out in the notes of the repository; here we only list them. There is no circularity with Theorem 5.1, whose proof uses only the estimates (H-a)–(H-e). The model $M(\alpha+2\delta,\alpha)$ with the Hurwitz parameters reproduces all 36 stored fits of Hurwitz lattices, and excludes the alternatives (Bessel order moved with β', no Szegő shift, lattice parameters) (script `m3a_hd_model`).

6.3. **Shifted Hurwitz lattices.** For $\lambda_k = (k+\delta)^p + \kappa$, $p>1$, $Z_\Lambda(0) = -\frac12 - \delta$ does not depend on κ, and $w\mapsto Z_\Lambda(-w)$ has the moved Weyl pole $q = 1-\frac1p$ with residue $R_q = -\kappa q/p$, and further poles at $q_m = m - \frac1p$, $m\ge2$, whose terms are $O(n^{-2p})$.

Theorem 6.2. *Under the hypotheses of Theorem D, with* $\beta' = \alpha+2\delta$,

$$\frac{n^p b_n}{c} = \sum_{0\le k<1+p} d_k(p,\beta',\alpha)\, n^{-k} - 2(2-p)\, c\,\kappa\, n^{-p} + O(n^{-1-p}),$$

and $-2(2-p)c\kappa = pR_qW(q)\pi\cot(\pi q)$. *The coefficients of* n^{-k}, $k<p$, *do not depend on* κ *and are those of Theorem 6.1.*

By Theorem 5.1 the integer part is the lattice's natural form in N_δ, and the pole term sits at the same natural index. For $1<p<2$ it lies between d_1/n and d_2/n^2; at $p=2$ its coefficient vanishes.

Outline. The index map is a Möbius change of variable, $t_\kappa = t \circ \zeta_\kappa$, $\zeta_\kappa(z) = z/(1-\kappa z/n^p)$. The Hurwitz outer scalar composed with ζ_κ has the right jumps but behaves like $C(-z)^{-1/2}$ at infinity: the exponent at infinity of an unnormalised scalar is not an invariant, the exponent of the normalising gauge at the hard edge is. With the gauge $G_\kappa = (-z)^{-\delta}\exp\mathcal{C}[\rho_1]$, $\rho_1 = t - t_\kappa$ and $\mathcal{C}$ a Cauchy transform on $(-\infty, 0)$, the normalised scalar tends to a constant at infinity and $G_\kappa = (-z)^{-\delta}E_\kappa(0)(1+O(n^{1-p}|z|^{1-1/p}))$ at the hard edge, so the Szegő shift is β', independent of κ. The jumps are those of $M(\beta', \alpha)$ with a band factor whose integer moments give the Toda parameters, and a branched part, the tilt $\sum_m 2\pi\cot(\pi q_m)a_m(z/n^p)^{q_m}$, $a_m = -R_{q_m}/q_m$, which comes together with the lip factor $e^{\pm 2\pi i\rho_1}$ from one function analytic across $(-\delta_0, 0)$, so the circle jumps stay meromorphic. The estimates transfer from (H-b)–(H-e). The first-order response of the model to the tilt z^q is the symbolic-j layer of Lemma 4.2(b) continued to $j = q$, so a tilt $e^{\tau x^q}$ contributes $-\frac{pqW(q)}{2}\tau(n/N_\delta)^p N_\delta^{-1-pq}$; for $\tau = 2\pi\cot(\pi q)a_1$ this is $pR_qW(q)\pi\cot(\pi q)(n/N_\delta)^p N_\delta^{-1-pq}$. □

The proof gives more: up to $O(n^{-2p})$, $n^p b_n/c$ is the full double series of $M(\beta', \alpha)$ with the Toda parameters of Λ, plus, for each pole, the first-order layer continued to $j = q_m$; it says nothing about products of pole terms. Fits at $(p, \delta, \kappa) = (2.5, -0.3, \pm 0.3)$ and $(1.5, -0.4, 0.4)$ confirm d_1, d_2 to 10^{-10} and $5\cdot 10^{-8}$ and the n^{-p} term to $4\cdot 10^{-7}$ (script `w_outer_e2e`). For $p < 1$ the route fails: the gauge density ρ_1 is of size $|\kappa|n^{(1-p)(2+p)/p}$ on the Bessel scale, so the lattice g-function cannot carry it.

7. Weyl-type spectra: Conjecture W

7.1. **The statement.** Let $\sigma := \sum_k w_k\delta_{1/\lambda_k}$ with:

- (W1) distinct nodes $0 < \lambda_1 < \lambda_2 < \cdots \to \infty$ (eigenvalues with multiplicity are merged into one node, and the multiplicity goes into the mass);
- (W2) an index function $\mathfrak{n}$ with $\mathfrak{n}(\lambda_k) = k$ and a termwise differentiable expansion $\mathfrak{n}(\lambda) = \lambda^{1/p} + \sum_{r\in\mathcal{R}} A_r\lambda^r + \mathfrak{n}_0 + O(\lambda^{-M})$, $\mathcal{R} \subset (-\infty, \frac{1}{p}) \setminus \{0\}$ finite for each M (this excludes fluctuating counting functions);
- (W3) masses $w_k = \lambda_k^{-s/p}v(\lambda_k)$, $v(\lambda) = v_0\lambda^\tau(1 + \sum_\rho b_\rho\lambda^{-\rho})$, $\rho > 0$, with $s_{\text{eff}} := s - p\tau > 1$ and, in this section, $\alpha := (s_{\text{eff}} - 1 - p)/p$;
- (W4) $p \neq 1$; (W5) no pole q below is an integer.

The spectral data are: $Z(w) = \sum_k \lambda_k^{-w}$, continued by (W2), with $Z(0) = \mathfrak{n}_0 - \frac{1}{2}$; the poles $q = -r$ of $Z(-w)$, $r \in \mathcal{R}$, with residues $R_q = qA_{-q}$ (the Weyl pole $-\frac{1}{p}$ excluded); the density factor $D(x) := \mathfrak{n}'(\lambda)/(\frac{1}{p}\lambda^{1/p-1})$ and the mass factor $V(x) := v(1/x)x^\tau/v_0$ at $x = 1/\lambda$, with $\log V + \log D =: \sum_\rho B_\rho x^\rho$; and the *Toda times*

$$t_j := -\frac{2Z(-j)}{j} + B_j \quad (j \in \mathbb{N}), \qquad t_\rho := B_\rho \quad (\rho \notin \mathbb{N}).$$

For the lattice, $Z(0) = -\frac{1}{2}$, $Z(-j) = \zeta(-jp)$, there are no poles, $D = V = 1$, and t_j are the times (5.3).

Conjecture 7.1. *Under* (W1)–(W5), *with* $\beta := \alpha - 2Z(0) - 1$, $\nu := \alpha$ *and* $N := n + \frac{\beta+1}{2} = n + \frac{\alpha}{2} - Z(0)$,

$$\frac{n^p b_n}{c} \simeq \sum_{\mathbf{r}} \mu_n^{\mathbf{r}}\mathfrak{D}_{\mathbf{r}}(n; \beta, \nu) + \sum_{q\notin\mathbb{N}} pR_qW(q)\pi\cot(\pi q)\Big(\frac{n}{N}\Big)^p N^{-1-pq} - \sum_{\rho\notin\mathbb{N}} \frac{p\rho W(\rho)}{2}\,t_\rho\Big(\frac{n}{N}\Big)^p N^{-1-p\rho},$$

with $\mu_{n,j} := -t_j n^{-jp}$ and the layers (5.1) of $M(\beta,\nu)$. The claim is: the integer and Toda layers to every order, each pole and fractional term to relative order $O(N^{-2})$; products of two pole or fractional terms, and of such a term with a Toda time, are not claimed.

The first sum is $M(\beta,\nu)$ flowed along the Toda hierarchy (Theorem 5.3); since $\varepsilon_j = pW(j)Z(-j) = -\frac{pjW(j)}{2}t_j$ at $B_j = 0$, there is one linear response $-\frac{p\rho W(\rho)}{2}t_\rho$ for every time, integer or not. At first order in each datum the conjecture is the dictionary (1.8) of the introduction. *Proved cases:* the lattice (Theorem 4.3, Corollary 5.2), Hurwitz lattices (Theorem 6.1), and the shifted Hurwitz lattices at first order in the poles (Theorem 6.2). *Exact case:* for $\lambda_k = (j_{\nu',k}/\pi)^2$, $s = 2$, the poles are half-integers, where $\cot(\pi q) = 0$, and the Toda times computed from the spectral data vanish to the precision of the regularised sums ($|t_1| < 10^{-21}$); so the conjecture predicts the bare model $M(\nu' - 1, -\frac{1}{2})$, which is Proposition 4.4. *Multiplicities:* for the sphere, nodes $l(l+1)$ with multiplicity $2l+1$, the distinct nodes have $p = 2$ (with multiplicities counted the Weyl exponent would be 1), and $Z(-j) = 0$ for every $j \geq 1$: with $t = l + \frac{1}{2}$, $(t^2 - \frac{1}{4})^j$ is even in t and $\zeta(-2m, \frac{3}{2}) = -2^{-2m}$, so $Z(-j) = -(\frac{1}{4} - \frac{1}{4})^j = 0$; mass and density factors cancel, and the conjecture predicts the bare model $M(1,0)$.

7.2. **Evidence fixed in advance.** For node sets with poles, with $Z(0) \neq -\frac{1}{2}$ and with masses that are not pure powers, every coefficient was predicted from the spectral data alone (regularised $Z(0)$, $Z(-j)$, residues, density and mass factors) and the recursion of §4.1 at 40 digits, and the predictions were stored before any of the nine measures was computed (the test script asserts the time stamps). The conjecture has no fitted constant. The recurrence coefficients, $n \leq 1600$, were computed in quadruple precision with a Gauss–Jacobi rule for the tail, and the coefficients obtained by least-squares fits of n-windows after subtraction of the exactly known terms. A coefficient passes if $|\text{fit} - \text{prediction}| \leq \max(3 \cdot \text{spread}, 2 \cdot 10^{-6}(1 + |\text{prediction}|))$; the alternatives are the conjecture with one piece switched off (no Szegő shift, no pole term, no $Z(-j)$, no density or mass tilt, no natural-index shift of the pole terms). Of 41 coefficients, 40 pass; the exception is the sphere at n^{-4}, outside the claimed precision. The families are (A) $\lambda_k = k^p + ak^{p-1}$, a Hurwitz offset a/p plus decaying corrections, $Z(0) = -\frac{1}{2} - \frac{a}{p}$, poles at $q_i = (i-1)/p$; (B) Bessel zeros $(j_{\nu',k}/\pi)^P$, $Z(0) = -\frac{\nu'}{2} - \frac{1}{4}$, poles at $(2i-1)/P$; (M2) nodes $l^{3/2}$ with masses $(2l+1)l^{-3}$, a fractional time $\rho = \frac{2}{3}$; (M1) the sphere, $s = 4$. Table 7.1 shows a selection.

The n^{-1} coefficients test the Szegő shift $\beta = \alpha - 2Z(0) - 1$ at $Z(0) \neq -\frac{1}{2}$: they agree to 10^{-8} or better, and $\beta = \alpha$ is excluded by 10^3–10^5 tolerances. For the Bessel anchor at $P = 2$ the bare model reproduces the coefficients $n^{-1}, \ldots, n^{-4}$ of the closed form to $2 \cdot 10^{-21}$. Before this test, the pole law alone (the second sum of the conjecture) had been checked on synthetic lattices $\lambda_k = (k + \gamma k^{-m})^p$ with the cotangent identified from one row and six further rows predicted in advance, which agree to $4 \cdot 10^{-5}$ or better in the windows $n \geq 200$ (scripts `spec_pole`, `spec_gamma`).

Remark 7.2 (the sphere at n^{-4} and a refinement). At n^{-4} the sphere misses $M(1,0)$ by $+0.23132$; the probes $k^2 \pm \frac{1}{4}$ (moved Weyl pole $q = \frac{1}{2}$, $R = \mp\frac{1}{16}$) miss by ∓ 0.23131, so the term is of first order in the pole, one order below its leading power. It is explained by a refinement found after this failure: each pole term carries the dressing of the zeta layer continued in the index, $[1 + g(q)N^{-2} + O(N^{-4})]$ with $g(w) = A(w) + B(w)\nu^2$ and A_j, B_j of Theorem A. Since B_j has the factor $1/(2j-1)$, g has a pole at $w = \frac{1}{2}$, where $\cot(\pi w)$ vanishes, and a pole at $q = \frac{1}{2}$ contributes the finite term $R_{1/2}W(\frac{1}{2})\pi^2(p+2)^2(p+4)(1-4\nu^2)n^{-3-p/2}/256$, which is $\frac{3\pi^2}{8}R_{1/2} = \mp 0.2313188$ for the probes (fits ∓ 0.2313059, spread $7 \cdot 10^{-4}$) and gives 7.7762407 for the sphere (fit 7.7762408). It vanishes at $\nu = \pm\frac{1}{2}$, which is why the Lommel measures show

measure	coeff.	prediction	fit (spread)	difference	next alternative
A: $p=1.5$, $a=-\frac{3}{4}$	n^{-2}	$+0.0984431$	$+0.0984431$ $(3\cdot10^{-5})$	$-3\cdot10^{-9}$	no pole: 1392
	$n^{-2.5}$	$+0.2215758$	$+0.2215749$ $(4\cdot10^{-5})$	$-9\cdot10^{-7}$	no $Z(-j)$: 1912
A: $p=3$, $a=-0.6$	n^{-2}	$+0.1347742$	$+0.1347742$ $(5\cdot10^{-9})$	$+5\cdot10^{-10}$	no pole: 31968
	n^{-3}	$+0.0452648$	$+0.0452649$ $(2.7\cdot10^{-6})$	$+1.0\cdot10^{-7}$	no shift: 1466
A: $p=0.6$, $a=-0.3$	n^{-2}	$+0.1329122$	$+0.1327116$ $(1.1\cdot10^{-3})$	$-2.0\cdot10^{-4}$	no pole: 27
B: $\nu'=0$, $P=1.5$	n^{-2}	$+0.1957151$	$+0.1957150$ $(1.4\cdot10^{-5})$	$-1.3\cdot10^{-7}$	no pole: 562
B: $\nu'=0$, $P=3$	n^{-2}	$+0.0742502$	$+0.0742502$ $(3\cdot10^{-9})$	$+5\cdot10^{-10}$	no pole: 10692
	n^{-3}	$+0.0049259$	$+0.0049264$ $(1.2\cdot10^{-5})$	$+4.8\cdot10^{-7}$	no shift: 276
M2	n^{-2}	$+0.1180556$	$+0.1180558$ $(3.8\cdot10^{-6})$	$+2.7\cdot10^{-7}$	no mass tilt: 21830
M1 (sphere)	n^{-3}	-5	-4.99999992 $(3\cdot10^{-6})$	$8\cdot10^{-8}$	one tilt only: 12851
	n^{-4}	$+7.5449219$	$+7.776241$ $(3.3\cdot10^{-4})$	$+0.23132$	not claimed

TABLE 7.1. Conjecture 7.1 against fits (scripts `w_predict`, `w_test`); the last column gives the miss of the closest alternative in units of the tolerance. The rows $p=0.6$ are the weakest tests. Fit-free residuals decay in all rows shown except the sphere's n^{-4} (script `w_resid`).

nothing. The refinement has one test made after it was formulated and before the measure was computed: for $k^2+\frac{1}{4}$ at $s=3.5$ it predicts $+2.9296023$ at n^{-4}, the fit gives $+2.9296226$ (spread $3.6\cdot10^{-3}$), and the unrefined value $+3.1030915$ is excluded (scripts `w_probe`, `w_probe2`). The refinement is a conjecture; Theorem 6.2 proves its premise (pole terms are the continued zeta layer) only at leading order. For branched mass tilts that do not come from a pole, the formal model has in addition a hard-edge companion term of relative order $N^{-2\rho-1}$, beyond the claimed precision for $\rho>\frac{1}{2}$ (all tests above); it is not proved for actual measures.

Remark 7.3 (further directions). The following are numerical or heuristic and are not used. *Drums* ($p=1$, outside the theorems): for smoothed Dirichlet spectra of planar domains, rescaled to unit Weyl density, the perimeter, a pole at $q=-\frac{1}{2}$ where W has a pole and $\cot(\pi q)=0$, appears at order $n^{-1/2}$ with the $0\cdot\infty$ limit $-\pi\beta/(2\sqrt{2})$ of the pole weight, $\beta=\mathcal{P}/\sqrt{4\pi\mathcal{A}}$, to 10^{-5} for four domains, and the corner and curvature constant enters through $pZ(0)$ (script `spec_corners`). *Anharmonic oscillators* $-\frac{\mathrm{d}^2}{\mathrm{d}q^2}+q^N$: the pole term is the rational number $\frac{N-1}{6(N+2)}$, conditionally on the second-order Bohr–Sommerfeld condition, and is reproduced to between 10^{-10} and $7\cdot10^{-8}$ (script `spec_homog`). *Irregular spectra*: if the nodes deviate from a smooth node function by a fraction of the spacing, the expansion is replaced by noise of relative size $n^{-1/2}$ (scripts `spec_jitter`, `spec_stats`). *Large order*: the integer layer F_p appears to be divergent, with factorially growing coefficients and a complex-conjugate pair of Borel singularities (scripts in `checks/large_order`).

7.3. **Open problems.**

Problem 7.4. (a) Conjecture 7.1 beyond the shifted Hurwitz family. *For a general Weyl-type node set the template of Theorem 6.1 needs an outer scalar built from a regularised determinant of Λ composed with an analytic index map, with normalising gauge exponent $-Z_\Lambda(0)-\frac{1}{2}$ at the hard edge; band-factor moments $Z_\Lambda(-j)/(jn^{jp})$; and the linear response of Lemma 2.3 for branched tilts. For $p<1$ the lattice g-function cannot carry the index offset (§6.3), and the n-dependent equilibrium of Proposition 3.1 is needed already for $\lambda_k=k^p+\kappa$. Products of pole terms are open even for the shifted Hurwitz lattices.*

(b) All layers in natural form. *Show that* $(N/n)^p n^p b_n/c$ *depends on* n *only through* N^{-2} *and the monomials* $X_j = \varepsilon_j N^{-1-jp}$. *Proved: the integer layer (Corollary 5.2), the leading coefficient of every layer (Theorem 5.4), and the instances of Theorem 4.3(ii). The shift of the layers of weight* ≥ 3 *would follow from the linear independence of* $\prod_j \zeta(-jp)^{r_j}$ *over* $\mathbb{Q}(p)$, *or from a version of the second-frame argument for the tilted measures of Theorem 5.3; the parity would follow from the evenness of the diagonal layer (Remark 5.5).*

(c) The second-order edge kernel. *Prove that the* N^{-2} *dressing of a pole term at* q *is the continuation* $g(q)$ *(Remark 7.2), or find the correct kernel; by the factor* $1/(2j-1)$ *it should combine* $(1-u)^{-1/2}$ *with* $(1-u)^{1/2}$. *The case* $p=1$, *where the two cotangents of Theorem 3.5 merge, is open.*

8. Relation to the literature

Spectral data and Jacobi parameters. Orthogonal polynomials whose moments are spectral zeta values at positive integers are classical for Bessel zeros: the Lommel polynomials are orthogonal on the reciprocal zeros [26] (Proposition 4.4), and an analogue for the zeros of J'_ν is given in [8]. The Tricomi–Carlitz polynomials, orthogonal on $\pm(k+\alpha)^{-1/2}$ [28], have explicit recurrence coefficients in which, from the recurrence as stated there, the offset enters only through the index $n+\alpha$, as in Theorem C (at $p=1$ in x^2, outside our range). Positive Jacobi matrices with compact inverse are studied in [25] through a regularised characteristic function; according to our reading, asymptotics of the Jacobi parameters are not treated there. We know of no expansion of recurrence coefficients in spectral zeta values at non-positive arguments, as in Theorems A–C. For continuous weights, singularities of the weight produce identifiable power-law or oscillating terms in the recurrence coefficients [19, 15]; the pole terms of Theorem D and Conjecture 7.1 may be viewed as an instance of this principle at the accumulation point of a discrete spectrum, with the poles of $Z(-w)$ as singularity data. Subleading corrections to Lanczos coefficients have recently been derived from endpoint data of continuous densities, including a hard-edge n^{-2} term [20], which we have not compared with f_2.

Constrained equilibrium. Existence, uniqueness and structure of equilibrium problems with an upper constraint are classical for finite constraints or compact intervals [21, 11]. For constraints of finite mass with compact support and no external field, the structure of Proposition 3.1 appears in the analysis of conjugate-gradient convergence [3]: according to [2, Lemma 1.5], which follows [3, Lemma 3.1(a)], a monotonicity criterion gives a single band and an endpoint equation that becomes (3.1) after a reflection, and according to [1, 4] the proof of [3, Theorem 2.1] gives a constrained analogue of the Buyarov–Rakhmanov formula [7]. In our setting that superposition is $\int_a^Y |F'(b)|\,\omega_{[0,b]}\,\mathrm{d}b$, which we have checked numerically, not proved, to agree with (3.2). What is new here, as far as we know, is the constraint of infinite mass at a hard edge with $Y=\infty$ allowed, where the finite-mass theory does not apply, the single-integral form (3.2) and the edge constant; Corollary 3.2 is an elementary evaluation, for the lattice problem of [22, Theorem 1.3]. We have not been able to consult [3] itself. We have not found Proposition 3.4; its termwise Mellin asymptotics is a standard technique.

Shifted index, reflection, Toda. For varying weights the recurrence coefficients have expansions in even powers of $1/n$ after a half-index shift, by a reindexing symmetry of the string equations [5], by the structure of the local parametrices [18], and as generating functions of maps [13]; for Jacobi-type weights see [17], and an exact pairing of $n+\alpha/2$ with α^2 occurs for semiclassical Laguerre weights [14]. Our reflection uses the grading of [18, Lemma 4.1]. We have not found the decoupling of the Szegő exponent from the Bessel order, the identity (5.2) or its proof from a second frame at a non-integer index. The Toda hierarchy for tilts is classical

[27]; re-expanding it in $1/n$ to generate expansion coefficients is the method of [13, Section 4] for varying weights; the mechanism of Theorem 5.4 is the genus-zero one of map enumeration [12, 13, 6], with endpoints as Riemann invariants of the dispersionless hierarchy [16]. The new ingredients there are the setting and the zeta-valued times, not the mechanism.

Hurwitz zeta values. Hurwitz's formula is quoted from [9, Theorem 1.1], where it is attributed to Hurwitz and quoted from Titchmarsh's monograph, which we have not consulted. We have not found the index transfer of Section 6 or Theorems C–D; the closest exact instances are the Lommel and Tricomi–Carlitz recurrences, in which an offset of the nodes enters only through the index.

Appendix A. Proof of Proposition 3.1

By (H2), $g(y) := \sqrt{y}\,\sigma(y) = y\sigma(y)/\sqrt{y}$ is strictly decreasing and $\sigma(y) \geq y_0\sigma(y_0)/y$ for $y < y_0$; if $Y = \infty$, (H3) gives $\int^\infty \sigma < \infty$ and $y\sigma(y) \to 0$. Write $U^\mu(z) = \int \log|z-x|^{-1}\,\mathrm{d}\mu(x)$.

The root. Substituting $y = au$, $F(a) = \int_1^{Y/a} a\sigma(au)(1-1/u)^{-1/2}\,\mathrm{d}u$. For fixed u, $a\sigma(au) = (au\,\sigma(au))/u$ is strictly decreasing in a by (H2), and the upper limit decreases, so F is strictly decreasing; on $[a_0, a_1] \subset (0, Y)$, $a\sigma(au) \leq a_0\sigma(a_0 u)$, so F is continuous by dominated convergence. As $a \to 0$, $F(a) \geq \int_a^Y \sigma \to \infty$. If $Y < \infty$, $F(a) \to 0$ as $a \to Y$. If $Y = \infty$, split at $2a$: $\int_{2a}^\infty \leq \sqrt{2}\int_{2a}^\infty \sigma \to 0$, and $\int_a^{2a} \leq a\sigma(a)\int_1^2 (1-1/u)^{-1/2}\,\mathrm{d}u \to 0$.

The density. Let $S(z) := \int_a^Y s(y)(z-y)^{-1}\,\mathrm{d}y$, $\tilde{G}(z) := ((z-a)/z)^{1/2}S(z)$ (principal branch) and $G(z) := \int (z-x)^{-1}\,\mathrm{d}\mu(x)$. For fixed $y \in (a, Y)$ put

$$k_y(z) := \frac{((z-a)/z)^{1/2} - \sqrt{(y-a)/y}}{z-y}, \qquad \rho_y(x) := \frac{1}{\pi}\sqrt{\frac{a-x}{x}}\,\frac{1}{y-x} > 0 \quad (0 < x < a).$$

The singularity of k_y at y is removable, so k_y is analytic off $[0, a]$, $O(|z|^{-1/2})$ at 0, bounded at a, and $\sim (1 - \sqrt{(y-a)/y})/z$ at infinity; it jumps by $-2\pi i\rho_y$ on $(0, a)$. By Liouville's theorem $k_y(z) = \int_0^a \rho_y(x)(z-x)^{-1}\,\mathrm{d}x$, and ρ_y has mass $1 - \sqrt{(y-a)/y}$. Since $\sigma = s\sqrt{(y-a)/y}$, Fubini's theorem (all terms positive, $|z-x|^{-1} \leq |\operatorname{Im} z|^{-1}$, $\int s(1-\sqrt{(y-a)/y}) \leq F(a)$) gives

$$\tilde{G}(z) = \int_a^Y s(y)\Big[\frac{\sqrt{(y-a)/y}}{z-y} + k_y(z)\Big]\,\mathrm{d}y = \int_a^Y \frac{\sigma(y)\,\mathrm{d}y}{z-y} + \int_0^a \frac{\nu(x)\,\mathrm{d}x}{z-x} = G(z),$$

because $\int s(y)\rho_y(x)\,\mathrm{d}y = \nu(x)$. Hence μ has mass $\int s = F(a) = 1$, $\operatorname{Re} G(x \pm i0) = 0$ on $(0, a)$, and $\nu > 0$. With $R(y) := \sqrt{y(y-a)}$, $R\sigma = s(y-a)$ gives $\tilde{G} = R(z)^{-1}[F(a) + \int \sigma R(z-y)^{-1}\,\mathrm{d}y]$, so also $\pi\sqrt{x(a-x)}\,\nu(x) = 1 - \int_a^Y \sigma(y)\sqrt{y(y-a)}(y-x)^{-1}\,\mathrm{d}y$ by (3.1). For the upper bound write $s = g\cdot(y-a)^{-1/2}$ and use $\int_a^\infty (y-a)^{-1/2}(y-x)^{-1}\,\mathrm{d}y = \pi(a-x)^{-1/2}$: $\nu(x) < g(a)/\sqrt{x} \leq g(x)/\sqrt{x} = \sigma(x)$.

The potential. On $(0, a)$, U^μ is constant because $\operatorname{Re} G = 0$. For $z \in (a, Y)$, $\varphi := U^\mu - \ell$ satisfies $\varphi(a) = 0$ and $\varphi'(z) = -((z-a)/z)^{1/2}H(z)$, $H(z) := \mathrm{PV}\int_a^Y s(y)(z-y)^{-1}\,\mathrm{d}y$, which exists since σ is locally Hölder. With $s = g(y-a)^{-1/2}$,

$$H(z) = \int_a^Y \frac{(g(y)-g(z))(y-a)^{-1/2}}{z-y}\,\mathrm{d}y + g(z)\,\mathrm{PV}\int_a^Y \frac{(y-a)^{-1/2}}{z-y}\,\mathrm{d}y;$$

the first integrand is positive since g decreases, and the principal value is 0 if $Y = \infty$ and $(z-a)^{-1/2}\log\frac{\sqrt{Y-a}+\sqrt{z-a}}{\sqrt{Y-a}-\sqrt{z-a}} > 0$ if $Y < \infty$. Hence $\varphi < 0$ on $(a, Y]$.

Uniqueness. Since $\log^+|x-y| \leq \log(1+x) + \log(1+y)$ and (H3) bounds the logarithmic moment of every $\mu' \leq \sigma$, $I(\mu') > -\infty$ for every competitor. Let $\mu' \leq \sigma$ have unit mass and $I(\mu') < \infty$, $\tau := \mu' - \mu$. Then $I(\mu') - I(\mu) = I(\tau) + 2\int (U^\mu - \ell)\,\mathrm{d}\tau$. The integrand vanishes

on $[0,a]$ (U^μ is continuous, the density of μ being at most $g(a)x^{-1/2}$ on $(0,a)$ and bounded on $[a,Y]$); on $(a,Y]$, $\tau\le 0$ and $U^\mu-\ell<0$, so the integral is ≥ 0. The signed measure τ has mass 0 and $\iint|\log|x-y||\,\mathrm{d}|\tau|\,\mathrm{d}|\tau|<\infty$ (the positive part by (H3), the negative part by the density bound and $I(\mu')<\infty$), so $I(\tau)\ge 0$ with equality only if $\tau=0$, by [22, Lemma 3.3]. □

Remark A.1. Under (3.1), $\nu(a^-)=\sigma(a)$, so μ is continuous at a; without (3.1), ν acquires a multiple of $(a-x)^{-1/2}$. If σ is Hölder of order $>\frac12$ near a, then $\sigma-\nu=C\sqrt{a-x}(1+o(1))$ with $C=-h(a)/(\pi\sqrt{a})>0$, $h(a):=\int_a^Y(g(y)-g(a))(y-a)^{-3/2}\,\mathrm{d}y-2g(a)(Y-a)^{-1/2}\mathbf{1}_{Y<\infty}$. The density and potential steps use only that g is decreasing; (H2) enters through the monotonicity of F. The hypotheses are not necessary but cannot simply be dropped: in a discretised energy minimisation, a constraint with a dip below the edge gives three roots of (3.1) and two bands. The root, the two forms of ν, the mass and the inequalities were checked by quadrature for the lattice at $p=0.6$, 1.5 and two further densities (script `m1_theoremQ`).

Appendix B. Closed forms and reproducibility

Closed forms of Lemma 4.2.

$$f_6=\frac{\phi_0+\phi_2\alpha^2+\phi_4\alpha^4+\phi_6\alpha^6}{(3p+2)^2(5p+2)}, \tag{B.1}$$

$$\begin{aligned}
\phi_0&=-\frac{p+1}{362880\,p^2}\big(450p^8+1590p^7-316p^6-23761p^5-66864p^4\\
&\qquad-65507p^3-18220p^2-1632p-40\big),\\
\phi_2&=\frac{(p+1)(p+2)(90p^6+270p^5+664p^4-165p^3-1649p^2-288p+28)}{11520\,p},\\
\phi_4&=-\frac{(p+1)(p+2)^2(20p^4+45p^3+162p^2+72p+16)}{1536},\\
\phi_6&=\frac{p(p+2)^3(16p^3+7p^2-4p-4)}{3072},
\end{aligned}$$

$$\begin{aligned}
g_{1,4}&=\frac{p+1}{5760\,p^2(3p+2)}\big(9p^6+117p^5+161p^4-569p^3-1550p^2-976p+48\big)\\
&\quad+\frac{(p+1)(p+2)(3p^4+33p^3+124p^2+182p+108)\alpha^2}{192\,p(3p+2)}+\frac{(p+2)^3(p^3-4p-4)\alpha^4}{128\,p(3p+2)},
\end{aligned} \tag{B.2}$$

$$h_2=\frac{(p+1)(p+2)(3p^2+2p+1)}{8p^2}+\frac{(p+2)(2p^3-p^2-7p-6)\alpha^2}{16p^2}. \tag{B.3}$$

At $(p,\alpha)=(2,-\frac12)$, $f_4=\frac{3}{128}$ and $f_6=\frac{5}{1024}$, as in Proposition 4.4.

Scripts. All scripts are in `research/G1/RH/spectral/checks` of the repository named in the data availability statement, next to their stored outputs. They use the project's toolkit (quadruple-precision recurrence coefficients, with caches keyed by a hash of the node array and of (p,s,K)), `mpmath` and `sympy`; scripts that test a prediction end with assertions and print "ALL n CHECKS PASSED" or "ALL IDENTITIES HOLD". The exact runs of Lemma 4.2 begin with the SHA-256 hashes of the scripts that produced them:

script	SHA-256	runs (content)
`m4_rh_exact.py`	`1e2904089fa3053cd6a7249dd4cd6149`	`m4_rh_exact_k4`, `_k6`
	`2be3bb9c90b46c0932ff2507c9d9ffcc`	(Lemma 4.2(a))
`m4_zeta_exact.py`	`3df62305bba0a4801a767c252eeee000`	`m4_zeta_exact`
	`31e499f7c80a36946542fea6cd834041`	((b), $j \le 5$; $g_{1,4}$)
`m4_rh_exact2.py`	`143a308b8744bfccf3110c3a3f4a60a8`	(shared engine)
	`f192268b0f65f78b64bf8bd76c116521`	
`m4_zeta_exact2.py`	`badda6a534fbf621c2bbef3eca6c7822`	`_A`: (b), j symbolic;
	`b26d13493a655a1655860449a8bab9f4`	`_B`: (c)

The first two runs also assert $b_k = a_k + (k - \frac{1}{2})a_{k-1}$ (the termwise derivative of [10, (10.40.2)]) and the recursion for u_k, v_k of [10, (9.7.2)] for $k \le 8$. The other scripts cited are:

used in	scripts
§3, App. A	`m1_theoremQ`, `m1_edge`, `spec_kernel`, `m2_thmK_plt1`
§4	`m4_rh4`, `m4_f8_denominator`, `spec_lommel`
§5	`m4_shift_exact`, `sigma_frame_exact` (exact), `sigma_frame_num`, `toda_measures`, `toda_layers`, `shift_toda`, `shift_exact`, `shift_dressing`, `m4_toda`, `m4_diag`
§6	`m3a_e37_decoupled`, `m3a_g5_exact` (exact), `m3a_hd_model`, `w_outer_e2e`
§7	`w_predict`, `w_test`, `w_resid`, `w_probe`, `w_probe2`, `spec_pole`, `spec_gamma`, `spec_corners`, `spec_homog`, `spec_jitter`, `spec_stats`, `large_order/*`

The shell script `run_all.sh` regenerates the exact runs (about 35 minutes) and the `m1_`, `m4_` and `spec_` computations; the scripts `sigma_*`, `toda_*`, `shift_*`, `m3a_*`, `w_*`, `m4_toda`, `m4_diag` and `large_order/*` are run individually, and the measures of §7.2 are cached outside the repository.

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Data and code availability

The scripts that produced the exact identities and the numbers quoted in this paper, with their outputs, are available in the repository
https://github.com/speagleg/jacobi-asymptotics; Appendix B lists them.

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